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(54) **AIR CONDITIONING SYSTEM USING A RESPONSIVE HYGROSCOPIC MATERIAL**

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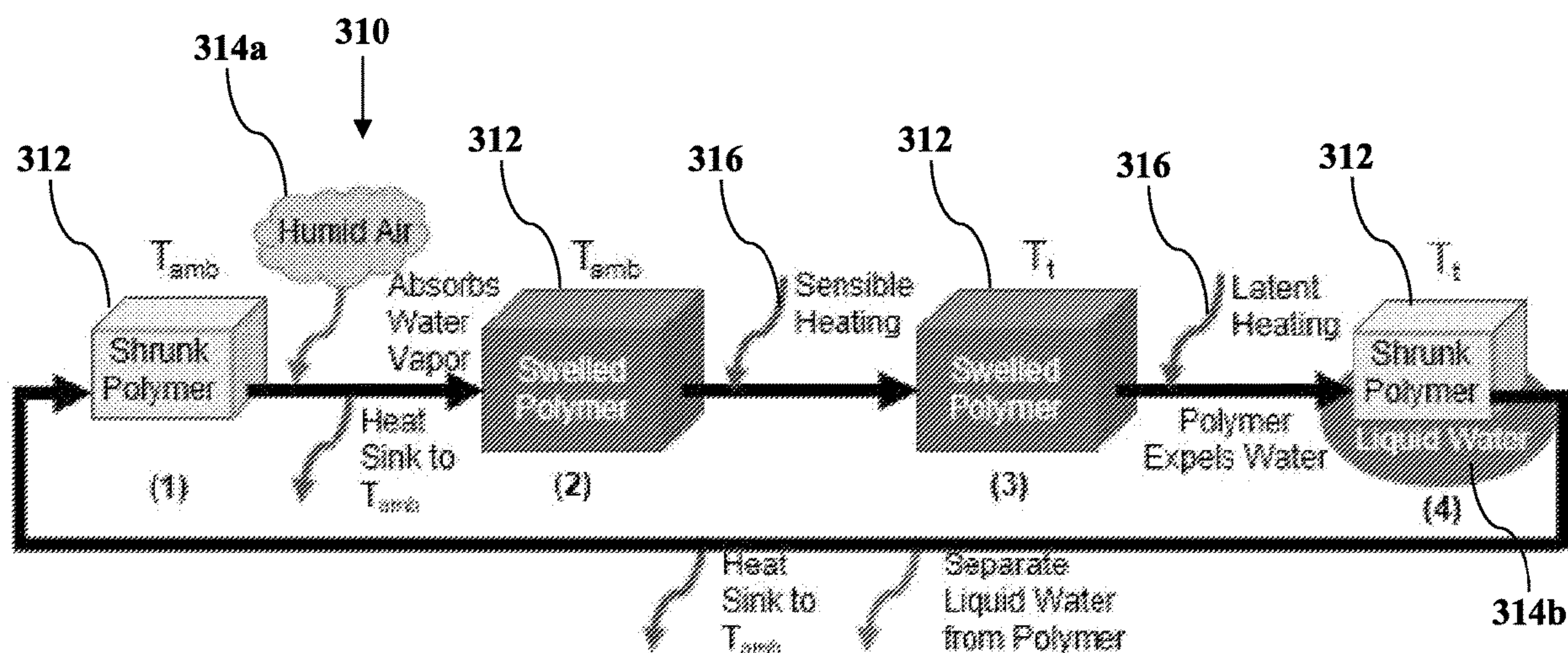
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CPC **F24F 3/1417** (2013.01)

(57) **ABSTRACT**

An exemplary embodiment of the present disclosure provides an air conditioning process. The air conditioning process can include a latent cooling stage including absorbing, via a responsive hygroscopic material, moisture from ambient air, heating, via a heat source, the responsive hygroscopic material above a transition temperature wherein the responsive hygroscopic material transitions from being hygroscopic to being hydrophobic, expelling, from the responsive hygroscopic material, liquid, and sensible cooling the responsive hygroscopic material below the transition temperature.

300



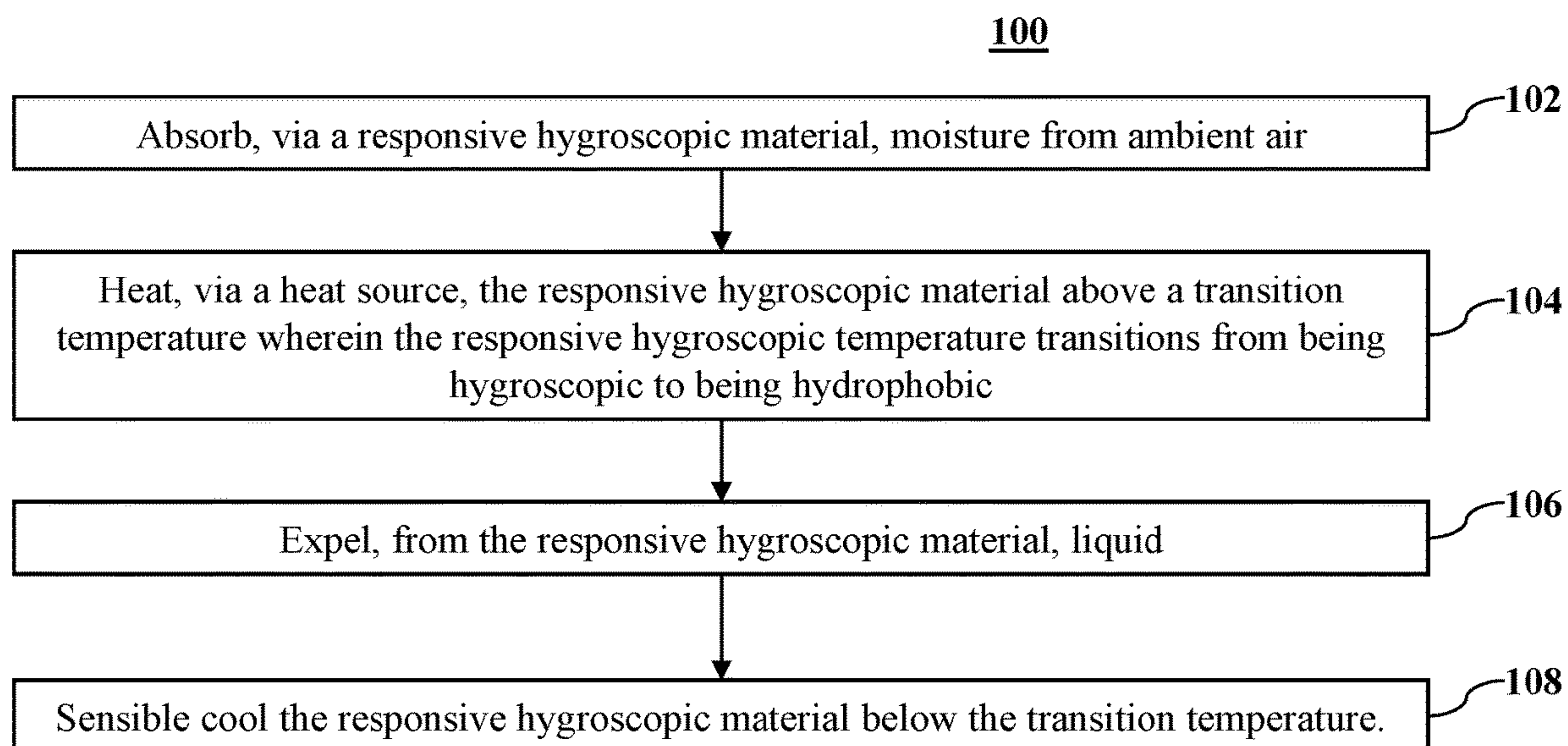


FIG. 1A

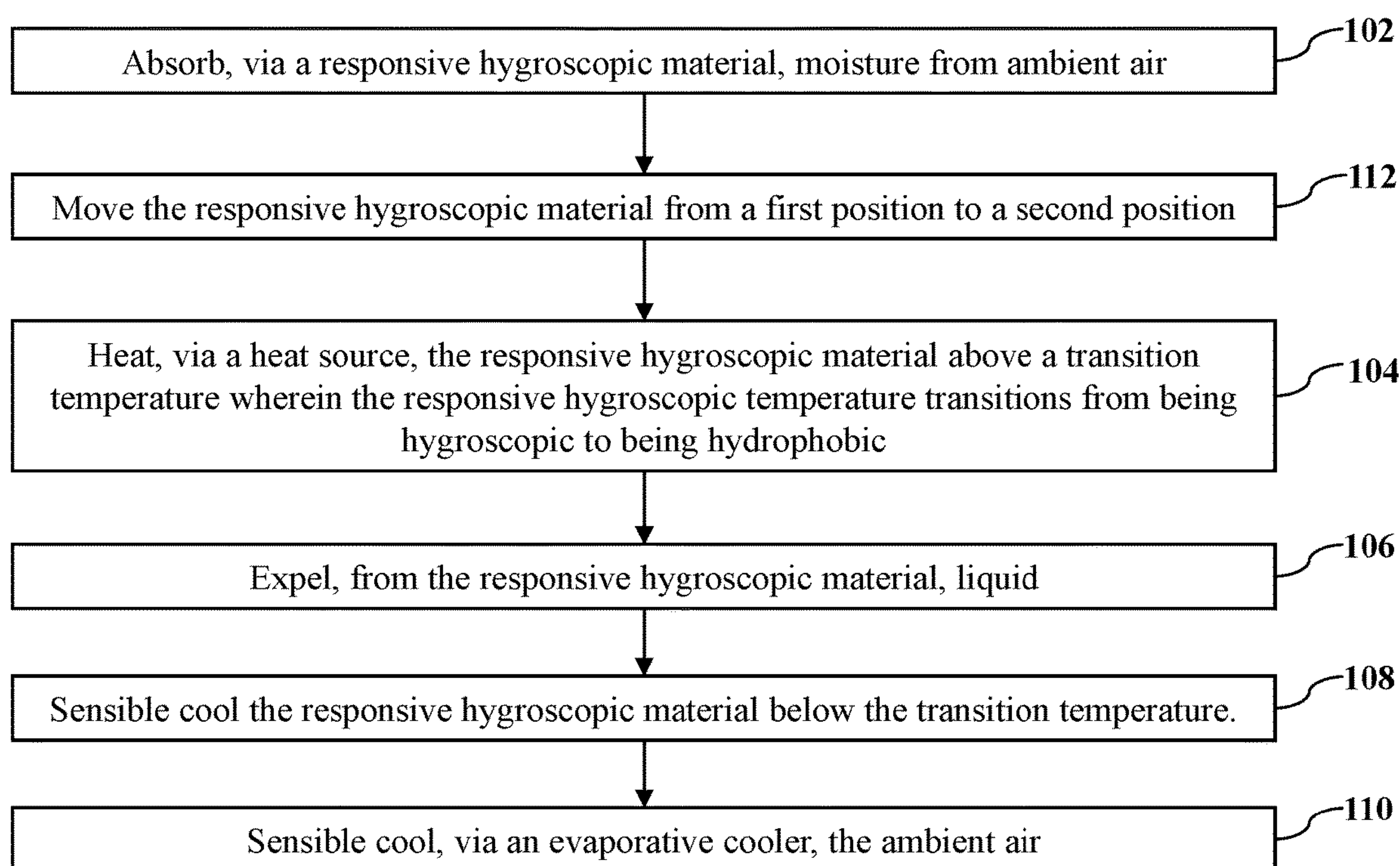
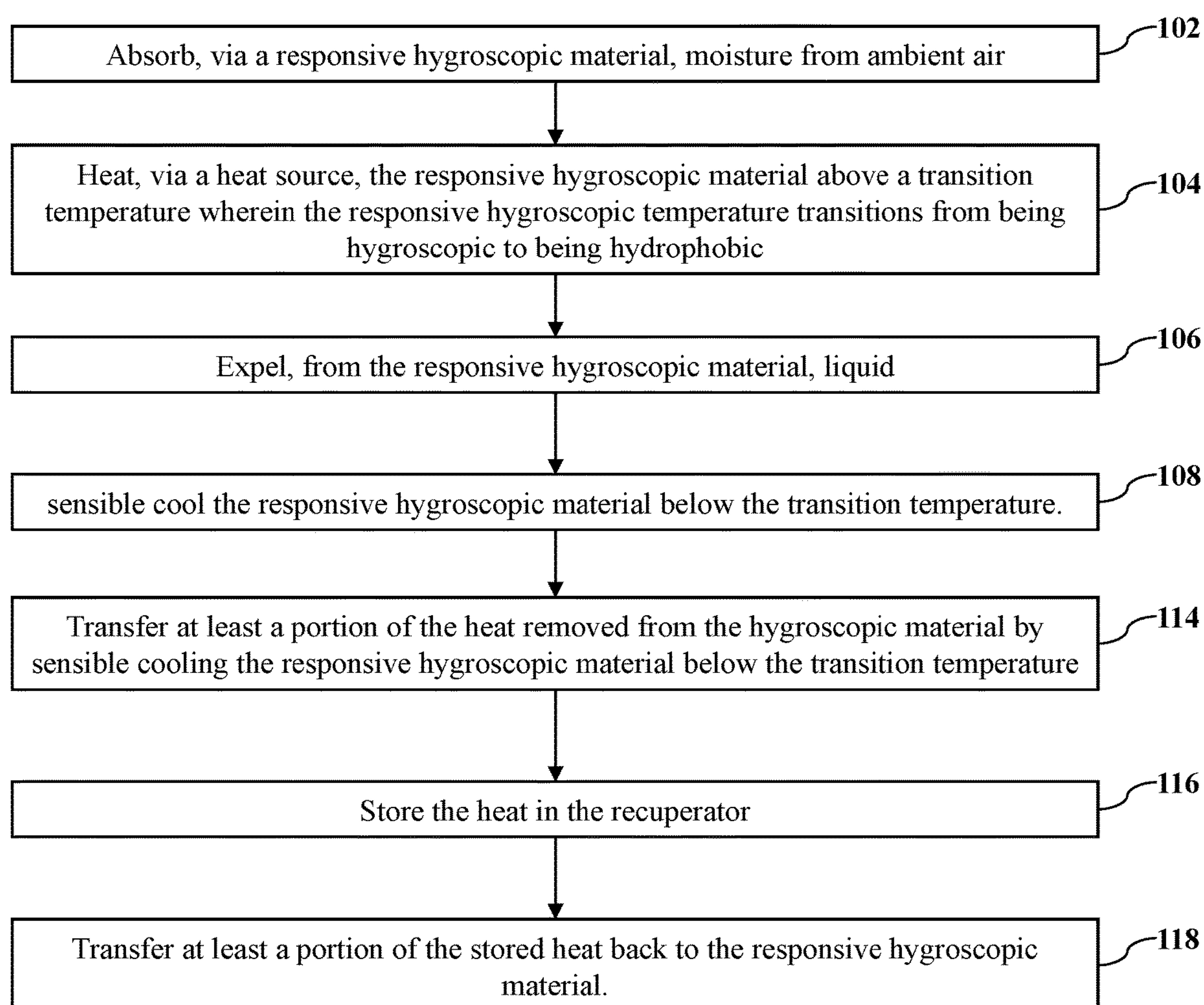


FIG. 1B

100**FIG. 2**

300

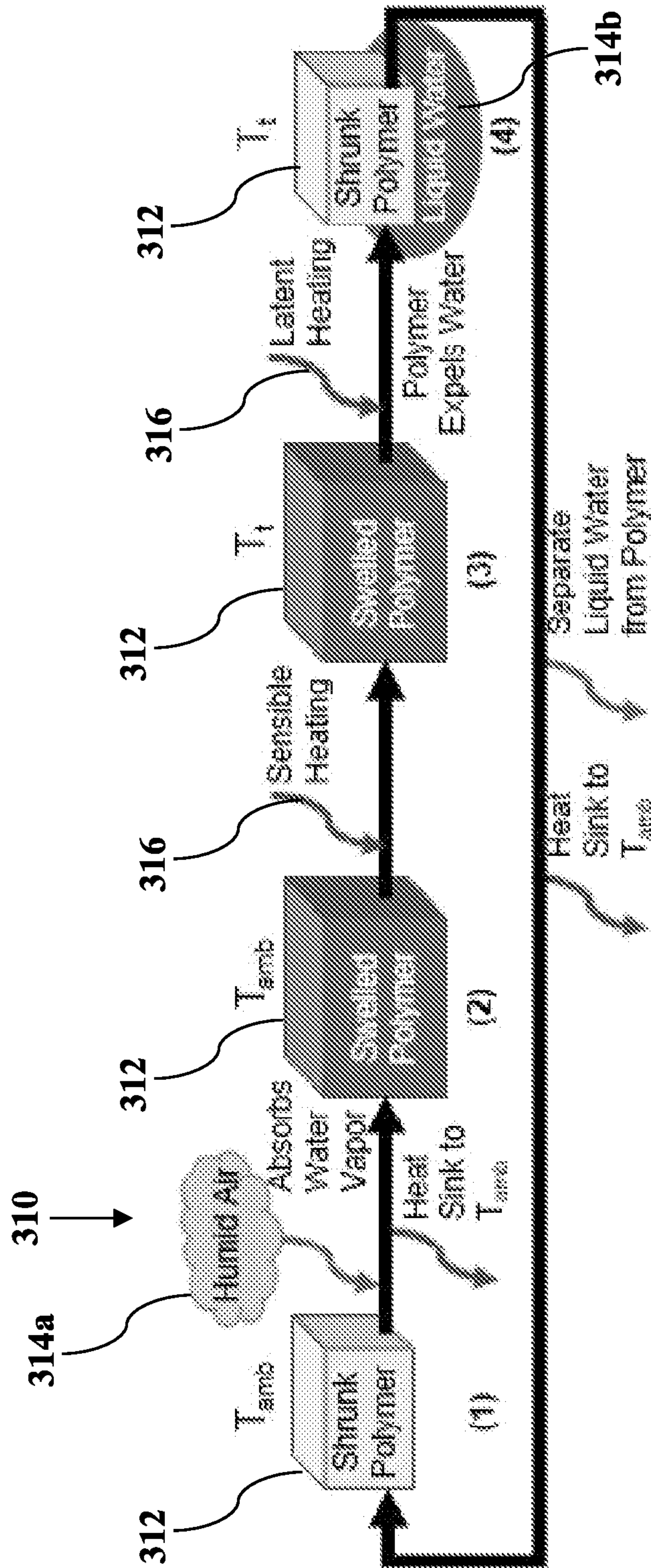


FIG. 3

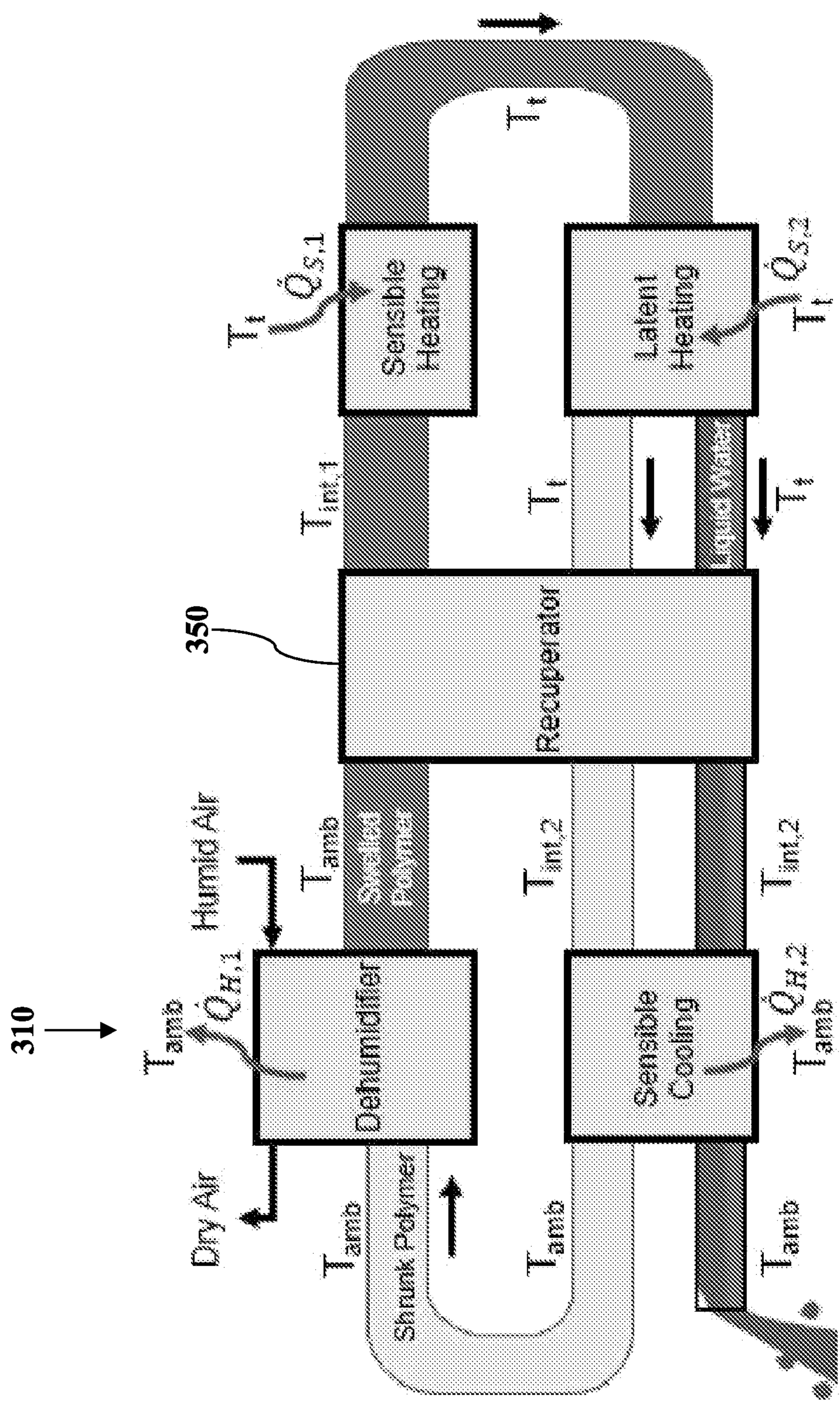


FIG. 4

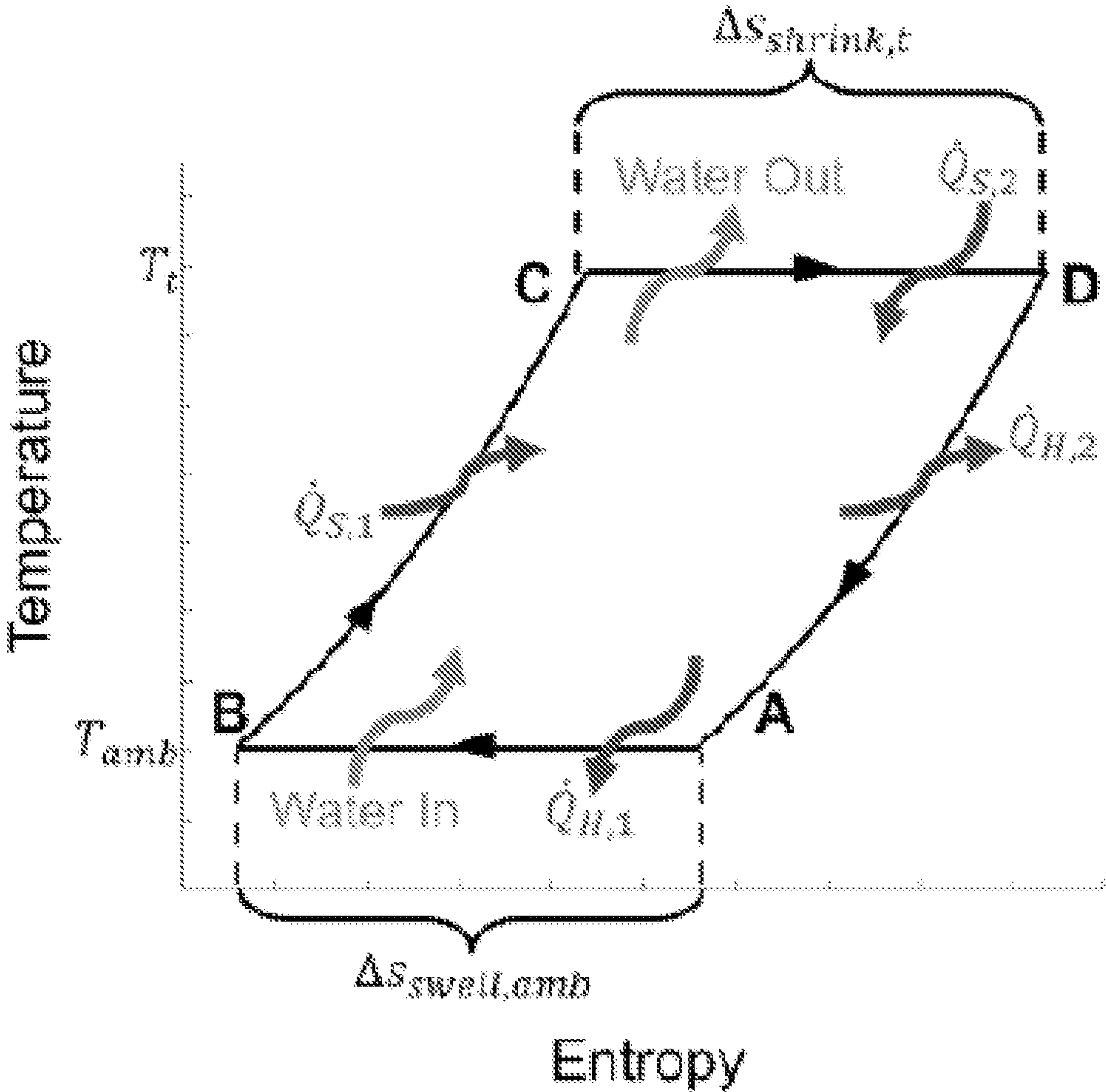
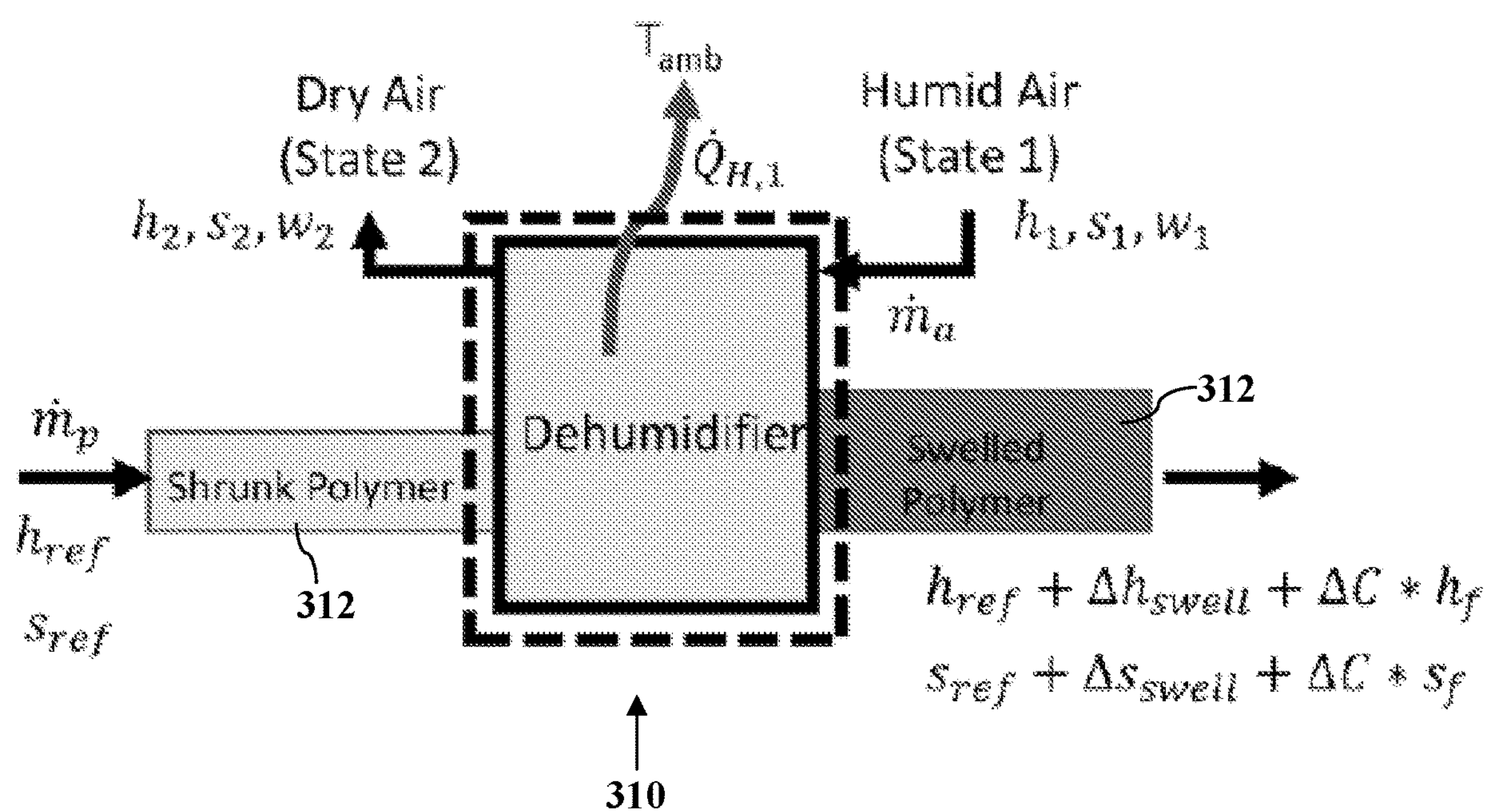


FIG. 5

$$T_{int,1} = T_{amb} + (T_t - T_{amb}) \quad (1)$$

**FIG. 6**

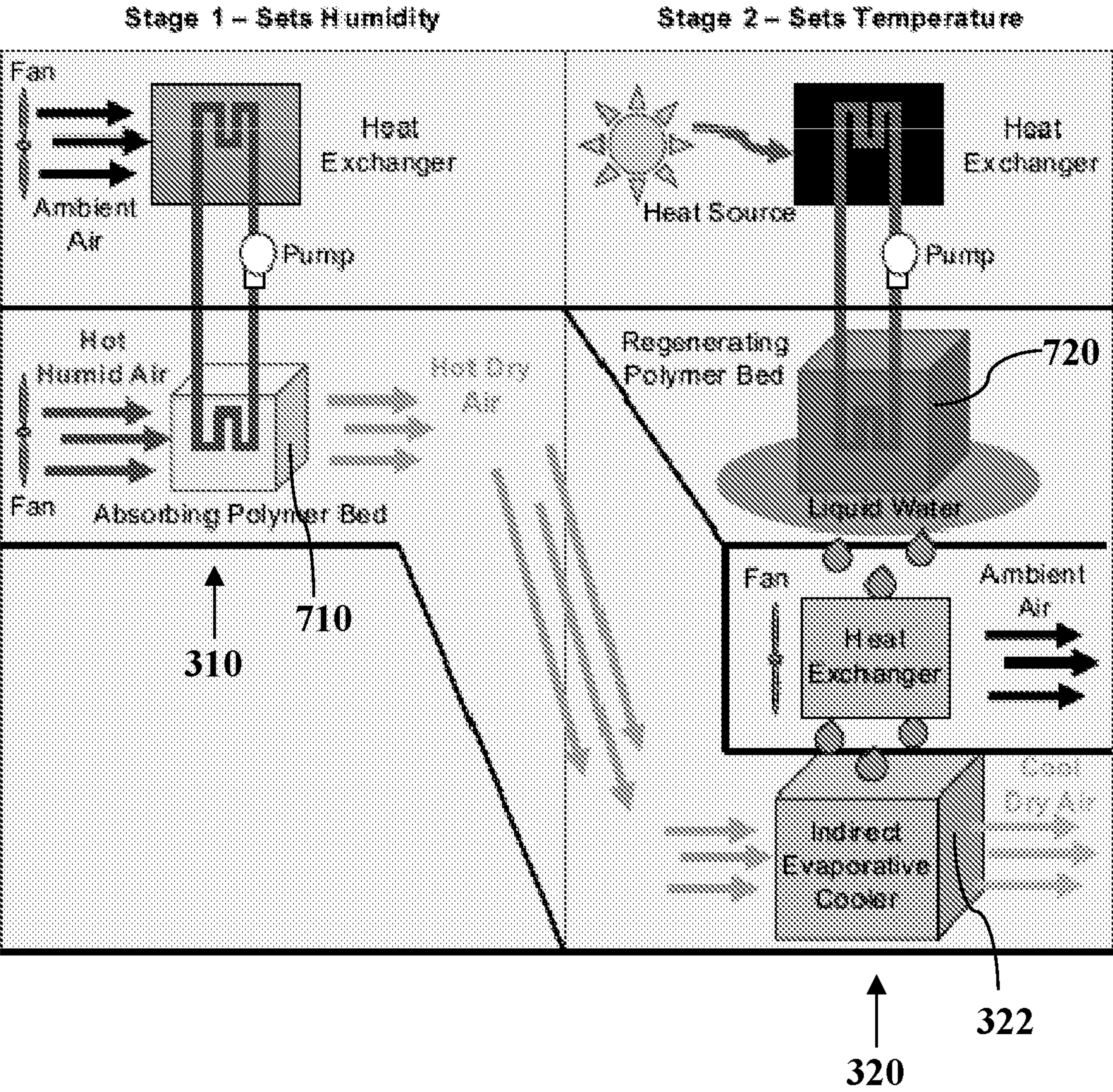


FIG. 7

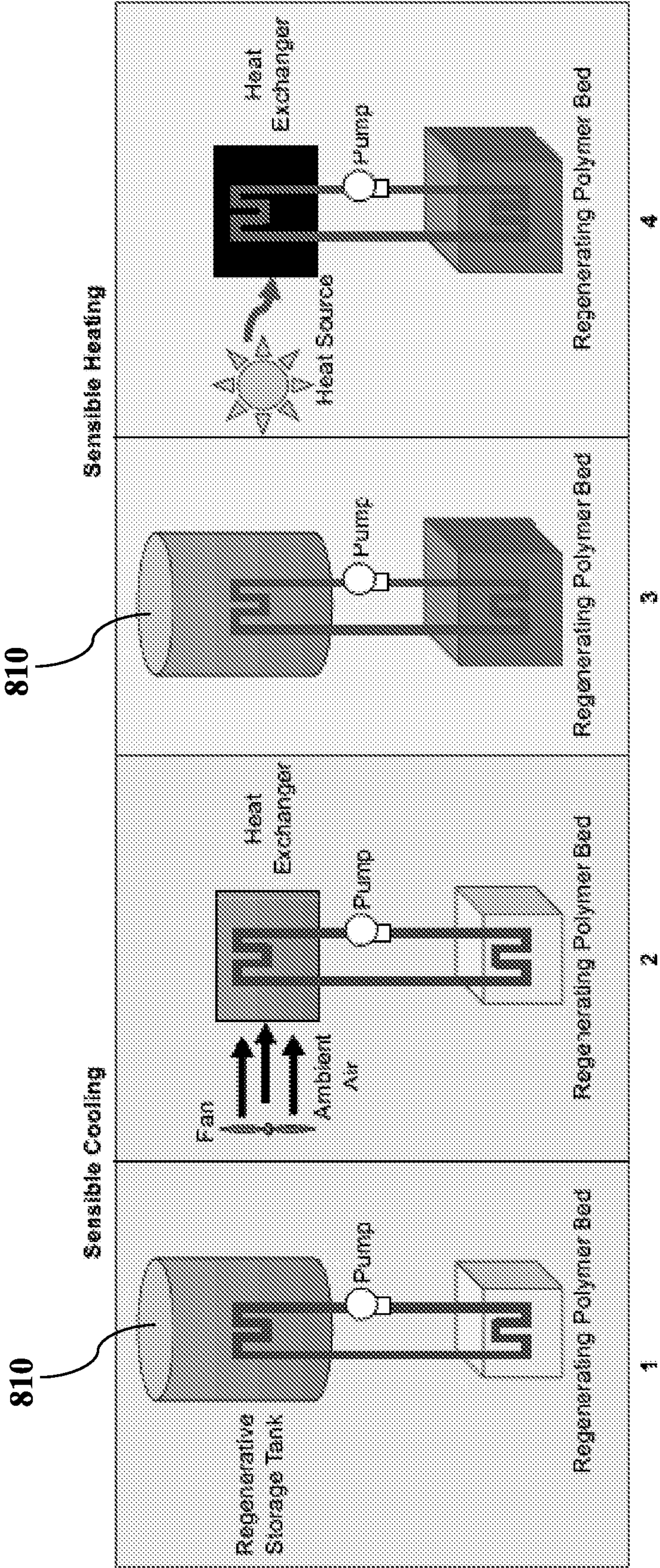
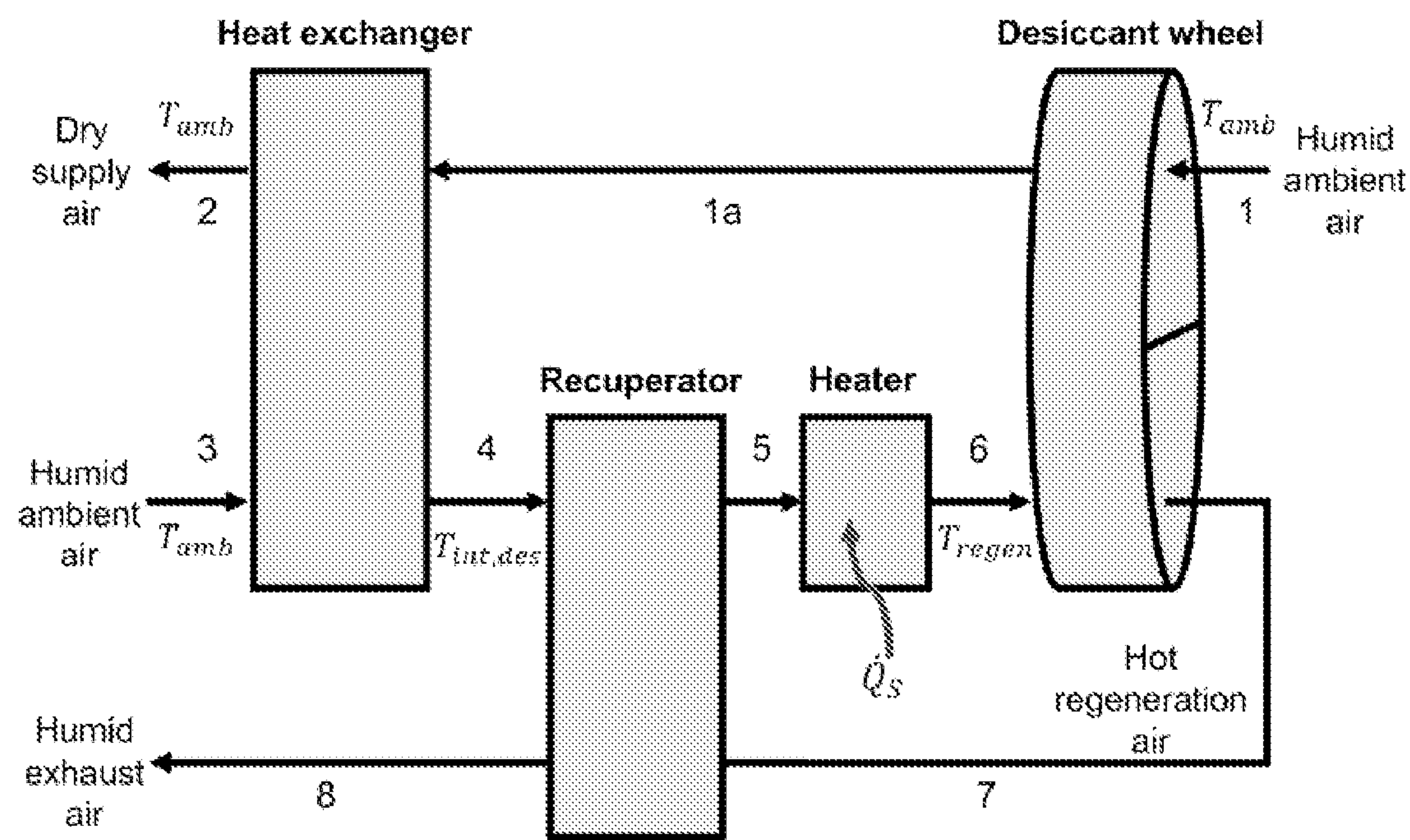


FIG. 8



PRIOR ART

FIG. 9

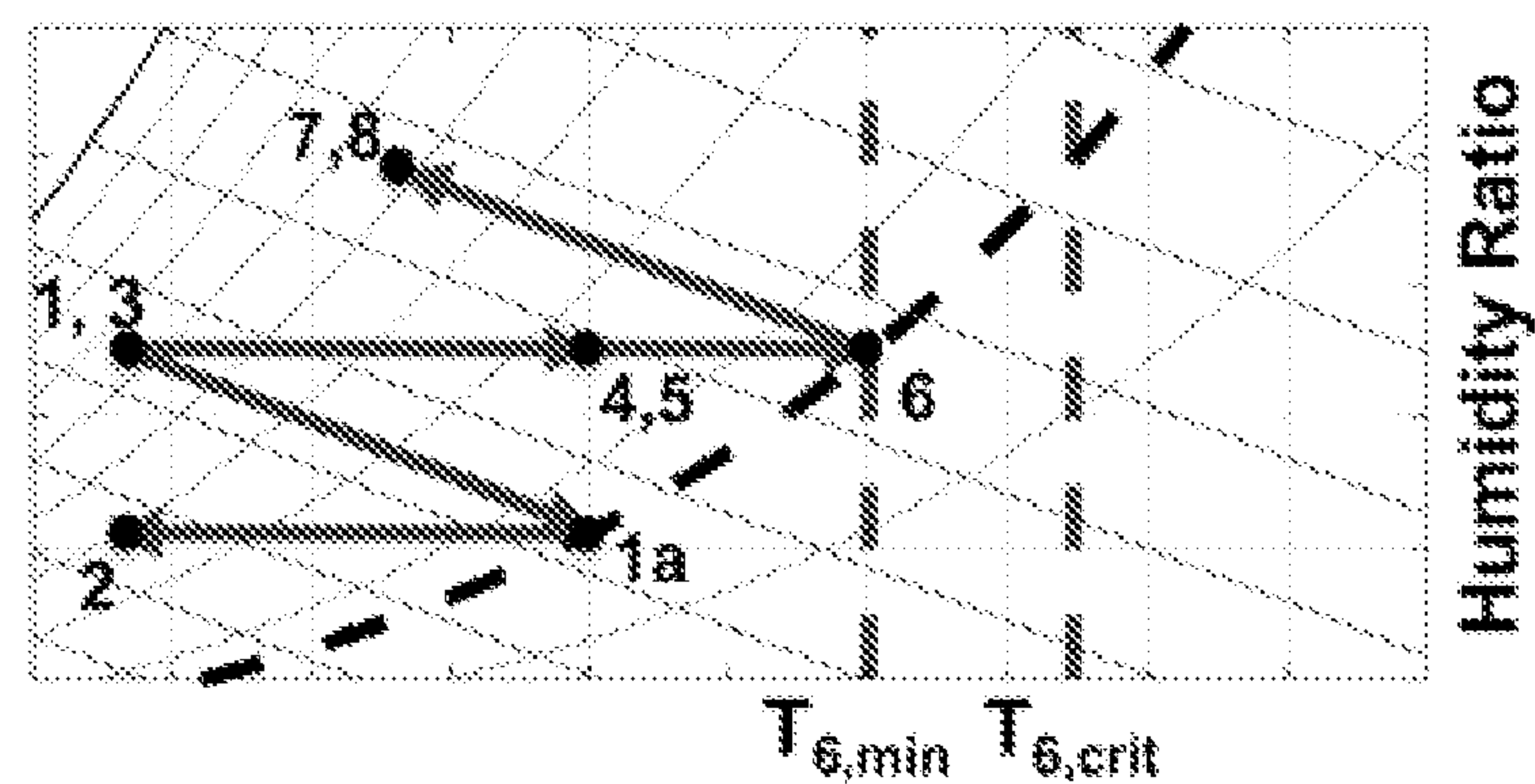


FIG. 10A

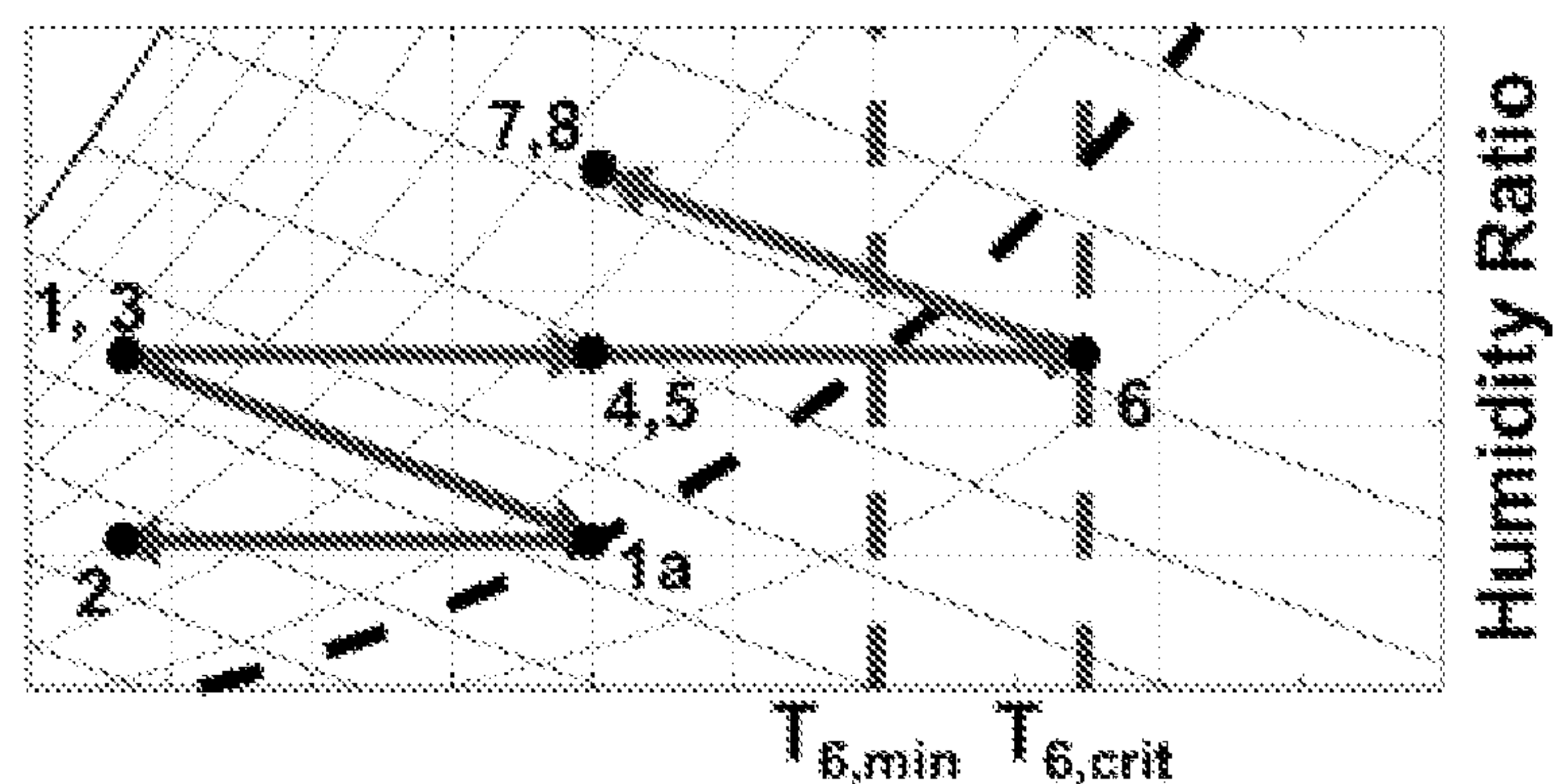


FIG. 10B

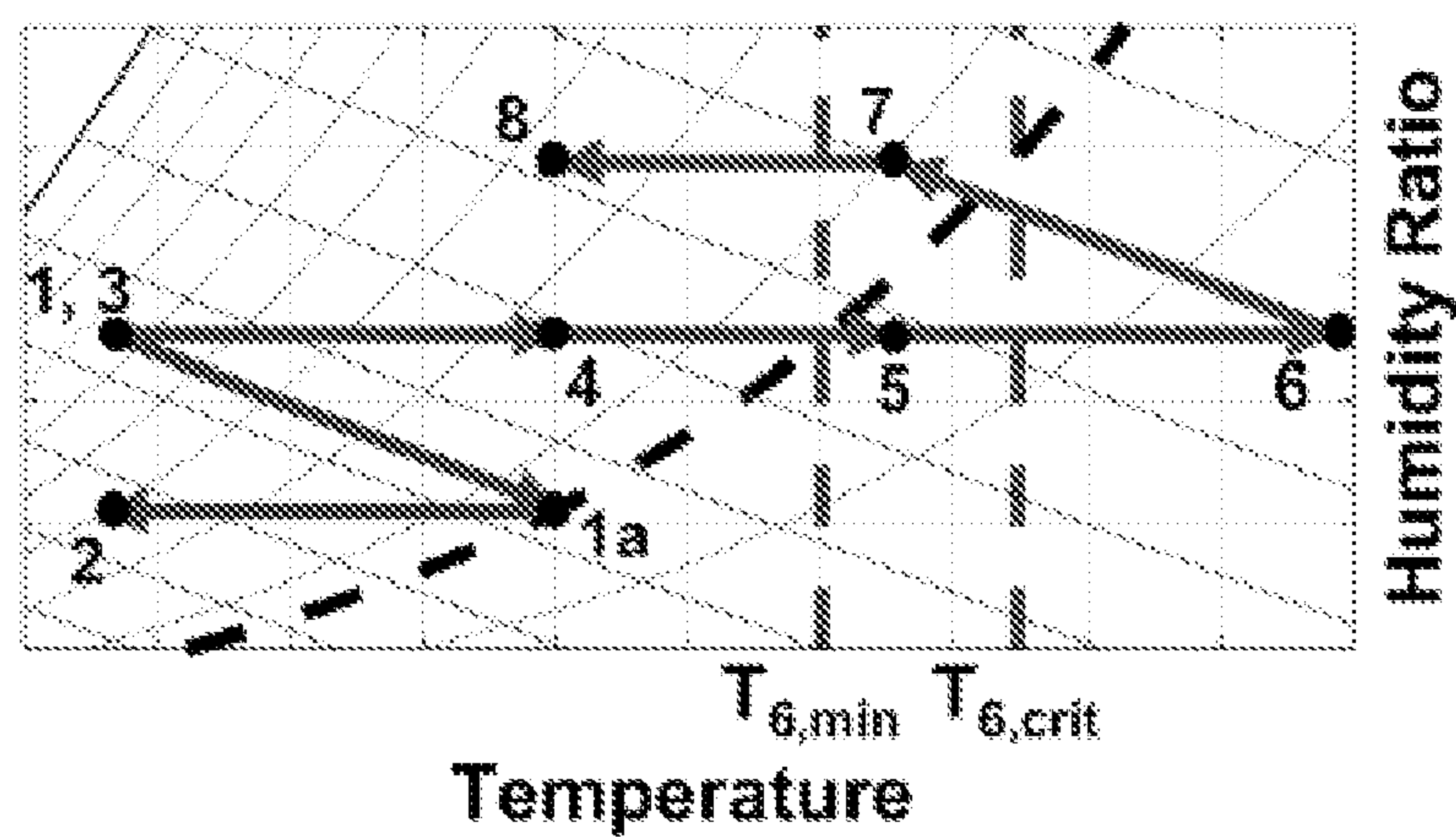


FIG. 10C

300

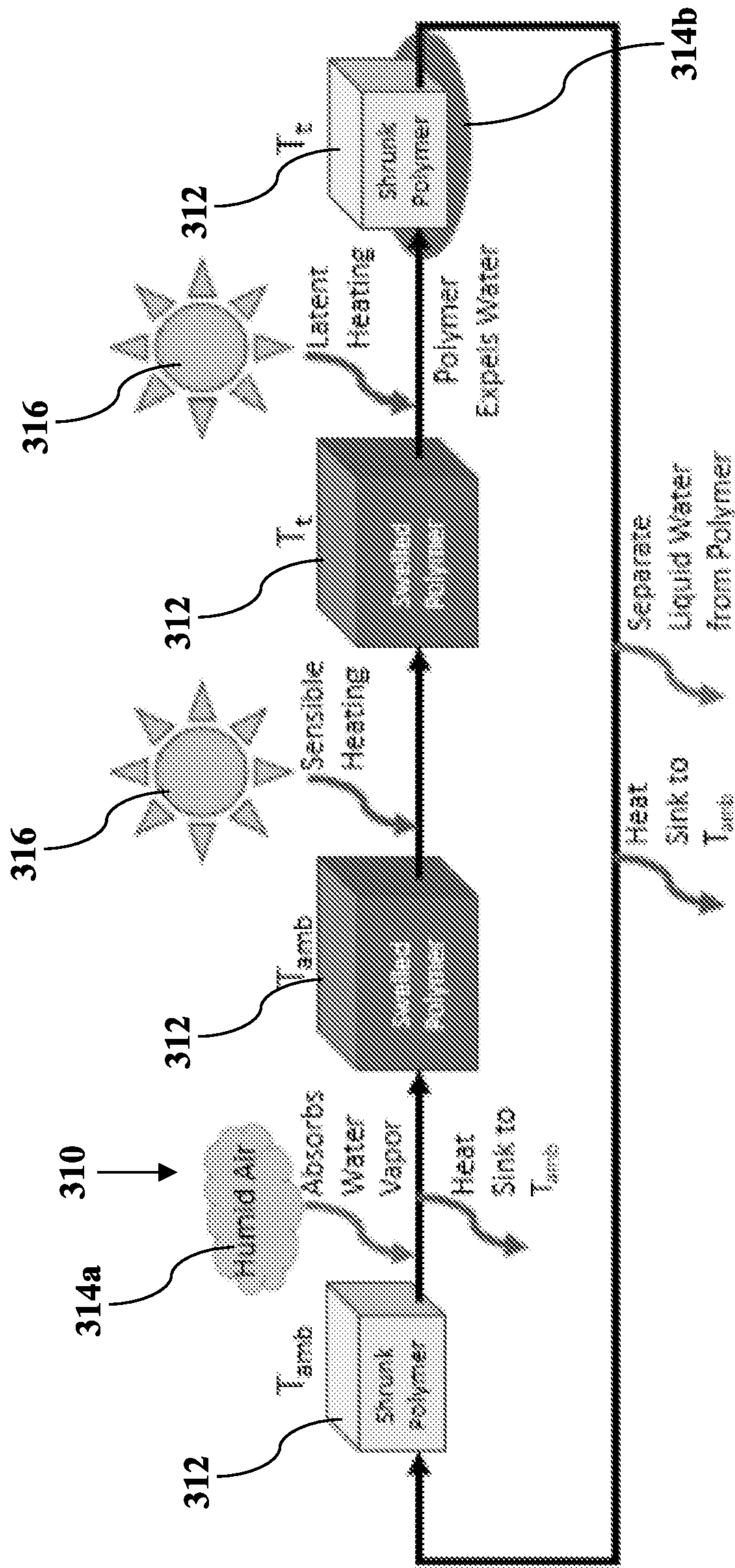


FIG. 11

310

320

Stage 1 – Sets Humidity

Stage 2 – Sets Temperature

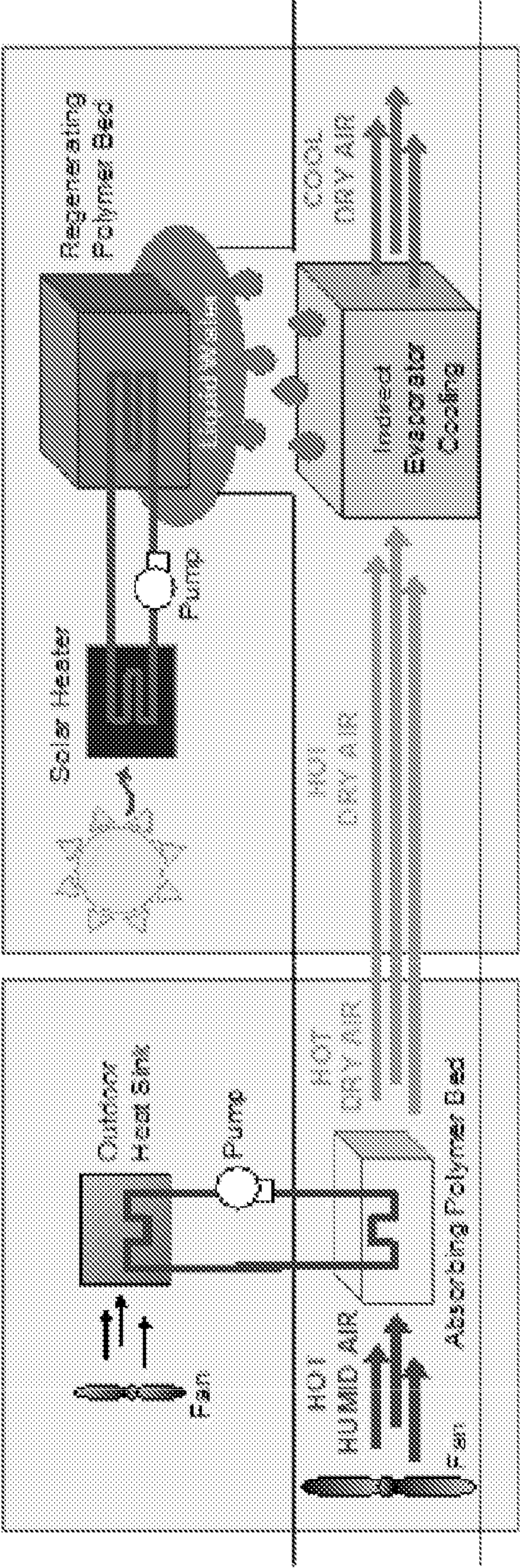


FIG. 12

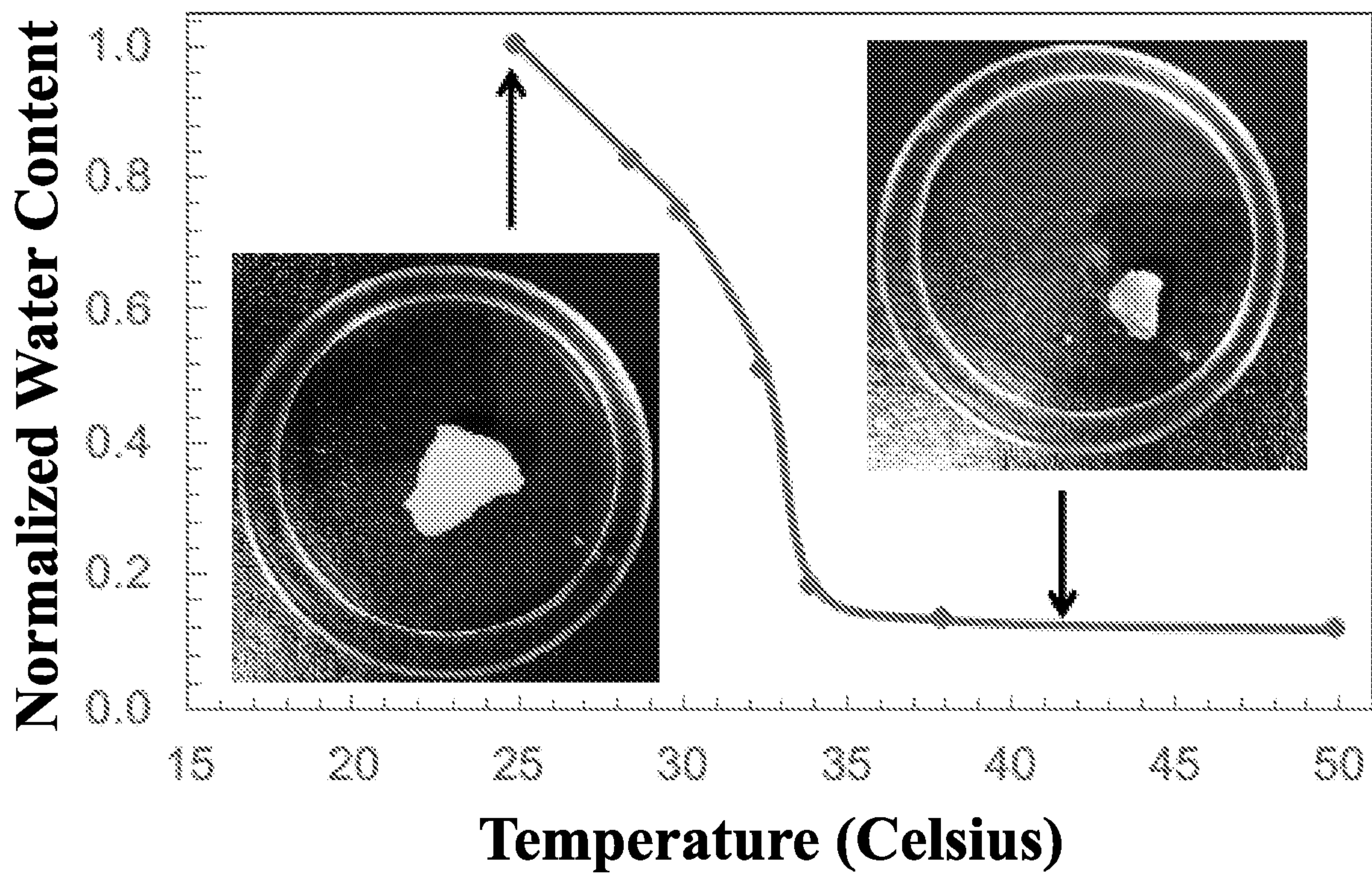


FIG. 13

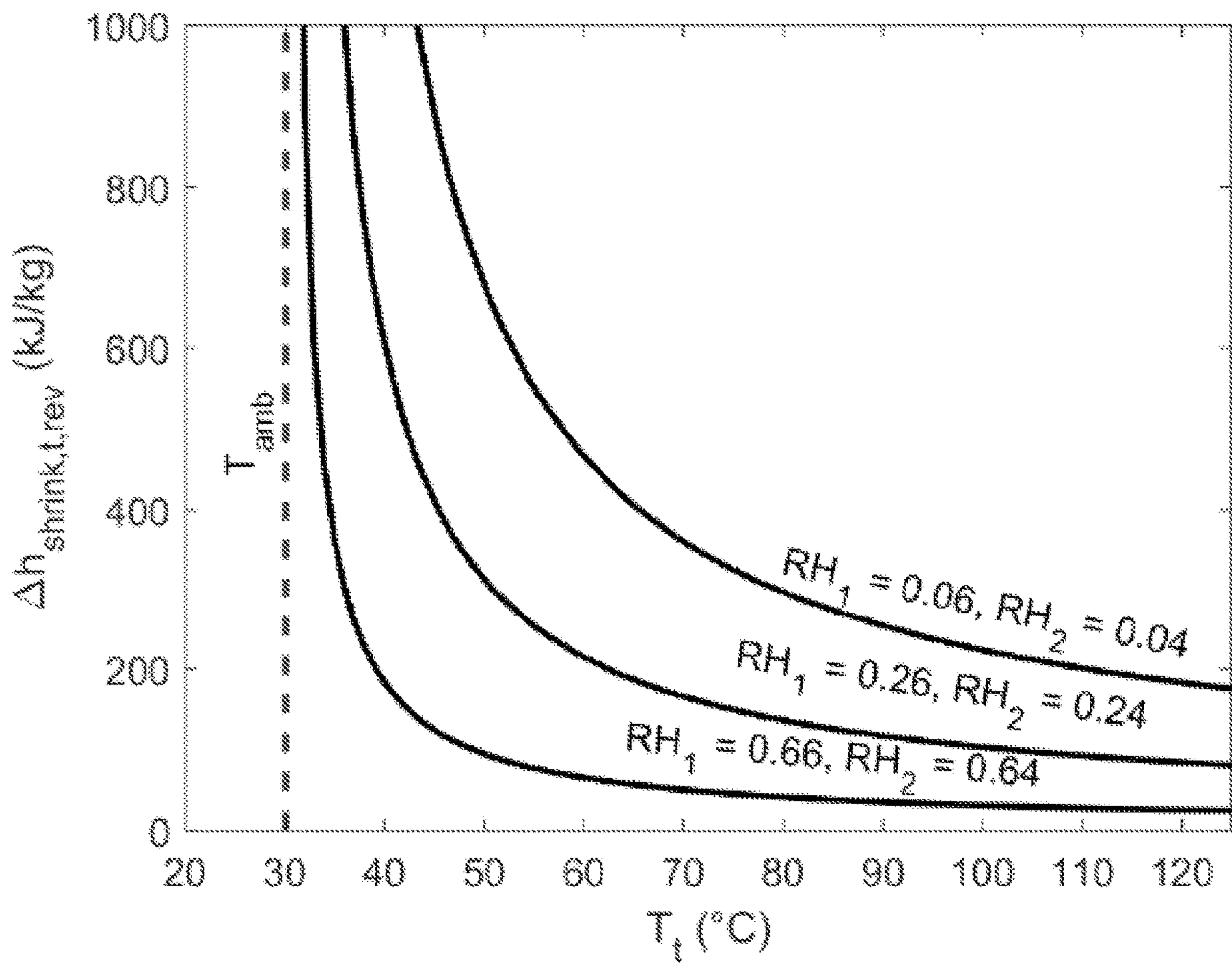


FIG. 14

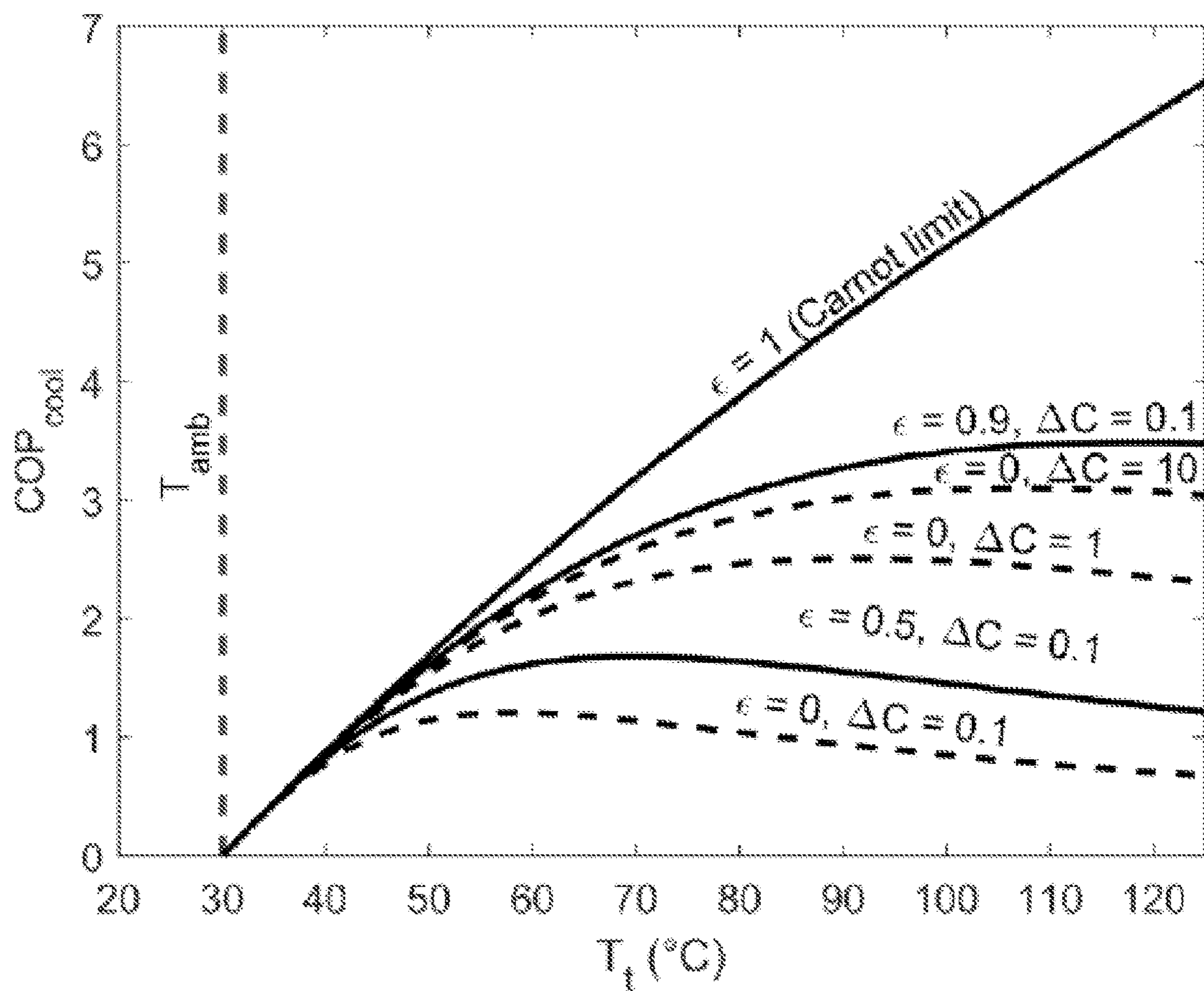


FIG. 15

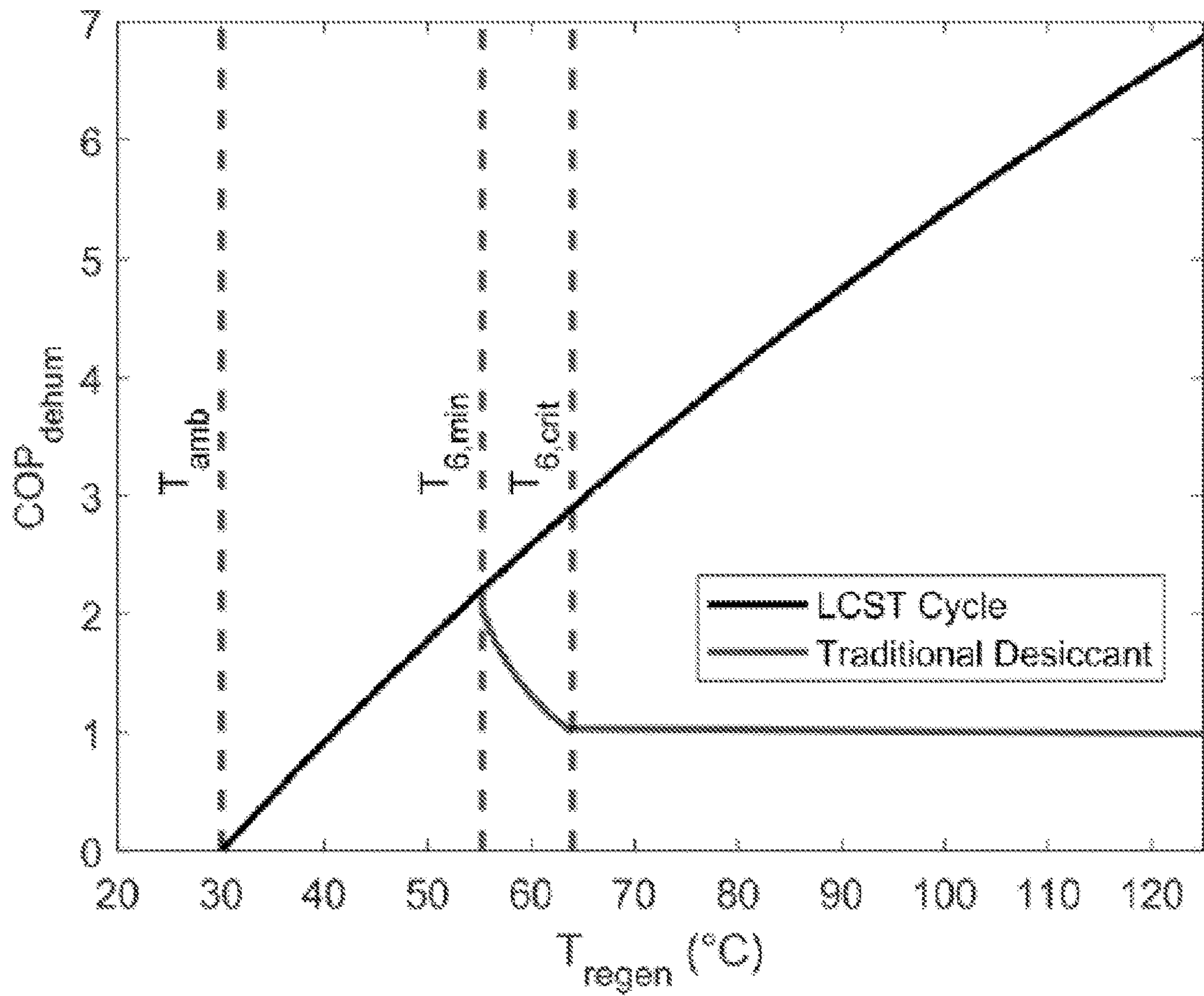


FIG. 16

AIR CONDITIONING SYSTEM USING A RESPONSIVE HYGROSCOPIC MATERIAL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application Ser. No. 63/378,262, filed on 4 Oct. 2022, which is incorporated herein by reference in its entirety as if fully set forth below.

FIELD OF THE DISCLOSURE

[0002] The various embodiments of the present disclosure relate generally to air conditioning systems, and more particularly to desiccant based air conditioning systems.

BACKGROUND

[0003] Current state-of-the-art air conditioning system is the vapor compression system. One of the primary operational deficiencies in the vapor compression system is that the same system is used to serve two independent functions: lowering the air temperature, which is known as the “sensible load”, and lowering the air humidity, which is known as the “latent load.” Vapor compression air conditioners use large amounts of energy to manage the latent load because these systems use an inefficient deep cooling process to decrease humidity. This deep cooling process can also cause water vapor from the air to freeze on the evaporator coils of the system, which leads to significant performance degradation.

[0004] Separate sensible and latent cooling (SSLC) air conditioning systems are a class of advanced air conditioning systems that use two separate sub-systems to manage the two independent functions of air conditioners. The first sub-system manages the latent load of the system by lowering the air’s humidity. The second sub-system manages the sensible load of the system by lowering the air’s temperatures. These systems improve occupant comfort by independently managing humidity and temperature. If designed properly, these systems can also be more energy efficient than vapor compression air conditioning systems. The first sub-system in SSLC systems typically uses a solid or liquid desiccant to manage humidity. These desiccants exhibit hygroscopic behavior, which means that they spontaneously absorb water vapor and thereby reduce the air’s humidity. The second sub-system in SSLC systems manages the temperature of the air. This second sub-system is typically a vapor-compression cooler, evaporative cooler, or some other cooling system.

[0005] The desiccants used in the first sub-system of SSLC systems eventually reach their water absorption capacity and need to be periodically regenerated. The regeneration process removes absorbed water from the desiccant and typically involves heating the desiccant to a high temperature. At this high temperature, water vapor desorbs from the desiccant and is rejected into a non-air-conditioned space. The regenerated desiccant can then again be used to absorb water vapor in the air-conditioned space and reduce humidity. Typical examples of solid desiccants are silica, activated carbon, calcium sulfate, calcium chloride, molecular sieves, zeolite, etc. Typical examples of liquid desiccants are aqueous solutions of lithium chloride, calcium chloride, etc.

[0006] There are several weaknesses that arise from using typical desiccants in air conditioning systems. First, typical desiccants require high temperatures to regenerate (typically greater than 100 degrees Celsius). These high regeneration temperatures eliminate the use of many cheap and inexpensive heat sources for the regeneration process. In addition, these high regeneration temperatures lead to parasitic energy loss and parasitic entropy generation that reduce the system’s efficiency. Second, typical desiccants release gaseous water during regeneration. The release of gaseous water during regeneration leads to the loss of water that could otherwise serve useful purposes (e.g., general water purposes and/or evaporative cooling). The invention described in this disclosure overcomes the weaknesses of the typical desiccant by using a responsive hygroscopic material as the desiccant.

BRIEF SUMMARY

[0007] The present disclosure relates to processes and systems for air conditioning. An exemplary embodiment of the present disclosure provides an air conditioning process. The air conditioning process can include a latent cooling stage including absorbing, via a responsive hygroscopic material, moisture from ambient air, heating, via a heat source, the responsive hygroscopic material above a transition temperature wherein the responsive hygroscopic material transitions from being hygroscopic to being hydrophobic, expelling, from the responsive hygroscopic material, liquid, and sensible cooling the responsive hygroscopic material below the transition temperature.

[0008] In any of the embodiments disclosed herein, the air conditioning process can further include a sensible cooling stage including sensible cooling, via an evaporative cooler, the ambient air. The evaporative cooler evaporates the liquid expelled from the responsive hygroscopic material.

[0009] In any of the embodiments disclosed herein, the evaporative cooler can be one of a direct evaporative cooler, indirect evaporative cooler, or a partially direct and partially indirect evaporative cooler.

[0010] In any of the embodiments disclosed herein, the latent and sensible cooling of the ambient air is performed without the use of conventional refrigerants.

[0011] In any of the embodiments disclosed herein, the process can further include moving the responsive hygroscopic material from a first position to a second position. Absorbing, via a responsive hygroscopic material, moisture from ambient air step can be performed at the first position. Heating, via a heat source, the responsive hygroscopic material above a transition temperature step can be performed at the second position.

[0012] In any of the embodiments disclosed herein, the process can further include transferring at least a portion of the heat removed from the hygroscopic material by sensible cooling the responsive hygroscopic material below the transition temperature, storing the heat in the recuperator, and transferring at least a portion of the stored heat back to the responsive hygroscopic material.

[0013] In any of the embodiments disclosed herein, the responsive hygroscopic material can include PNIPAAm and one or more hydrogels.

[0014] In any of the embodiments disclosed herein, the heat source can include one or more of solar heat, waste heat, gas heat, or electric heat.

[0015] An exemplary embodiment of the present disclosure provides an air conditioning system. The system can include a latent cooling stage including a responsive hygroscopic material configured to absorb moisture from ambient air, a heat source, and a heat exchanger configured for sensible cooling the responsive hygroscopic material. The heat source can be configured to heat the responsive hygroscopic material above a transition temperature wherein the responsive hygroscopic material transitions from being hygroscopic to being hydrophobic causing the responsive hygroscopic material to expel a liquid previously absorbed from the ambient air.

[0016] In any of the embodiments disclosed herein, the air conditioning system can further include a sensible cooling stage including an evaporative cooler configured for sensible cooling the ambient by evaporating the liquid expelled from the responsive hygroscopic material.

[0017] In any of the embodiments disclosed herein, the evaporative cooler can be a direct evaporative cooler, an indirect evaporative cooler, or a partially direct and partially indirect evaporative cooler.

[0018] In any of the embodiments disclosed herein, the latent and sensible cooling of the ambient air is performed without the use of conventional refrigerants.

[0019] In any of the embodiments disclosed herein, the air conditioning system an actuator configured to move the responsive hygroscopic material from a first position to a second position. At the first position, the responsive hygroscopic material can absorb moisture from the ambient air. At the second position, the heat source can heat the responsive hygroscopic material above the transition temperature.

[0020] In any of the embodiments disclosed herein, the air conditioning system can further include a first polymer bed. The first polymer bed can include the responsive hygroscopic material. The air conditioning system can include a second polymer bed. The second polymer bed can include the responsive hygroscopic material. The air conditioning system can include one or more actuators which can be configured to change the airflow, as between inside air and outside air, passing over each of the first polymer bed and second polymer bed.

[0021] In any of the embodiments disclosed herein, the air conditioning system can further include a recuperator. The recuperator can be configured to transfer at least a portion of the heat removed from the hygroscopic material by sensible cooling the responsive hygroscopic material below the transition temperature, storing the heat in the recuperator, and transferring at least a portion of the stored heat back to the responsive hygroscopic material.

[0022] In any of the embodiments disclosed herein, the responsive hygroscopic material can include PNIPAAm and one or more hydrogels.

[0023] In any of the embodiments disclosed herein, the heat exchanger can include an ambient heat exchanger.

[0024] An exemplary embodiment of the present disclosure provides an air conditioning system. The system can include a latent cooling stage and a sensible cooling stage. The latent cooling stage can include a responsive hygroscopic material configured to absorb moisture from ambient air, a heat source configured to heat the responsive hygroscopic material above a transition temperature wherein the responsive hygroscopic material transitions from being hygroscopic to being hydrophobic causing the responsive hygroscopic material to expel a liquid previously absorbed

from the ambient air, and a heat exchanger configured for sensible cooling the responsive hygroscopic material. The sensible cooling stage can include an evaporative cooler configured for sensible cooling the ambient by evaporating the liquid expelled from the responsive hygroscopic material.

[0025] In any of the embodiments disclosed herein, the air conditioning system can include a recuperator configured to transfer at least a portion of the heat removed from the hygroscopic material. Transferring at least a portion of the heat removed from the hygroscopic material can be done by sensible cooling the responsive hygroscopic material below the transition temperature, storing the heat in the recuperator, and transferring at least a portion of the stored heat back to the responsive hygroscopic material.

[0026] These and other aspects of the present disclosure are described in the Detailed Description below and the accompanying drawings. Other aspects and features of embodiments will become apparent to those of ordinary skill in the art upon reviewing the following description of specific, exemplary embodiments in concert with the drawings. While features of the present disclosure may be discussed relative to certain embodiments and figures, all embodiments of the present disclosure can include one or more of the features discussed herein. Further, while one or more embodiments may be discussed as having certain advantageous features, one or more of such features may also be used with the various embodiments discussed herein. In similar fashion, while exemplary embodiments may be discussed below as device, system, or method embodiments, it is to be understood that such exemplary embodiments can be implemented in various devices, systems, and methods of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The following detailed description of specific embodiments of the disclosure will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the disclosure, specific embodiments are shown in the drawings. It should be understood, however, that the disclosure is not limited to the precise arrangements and instrumentalities of the embodiments shown in the drawings.

[0028] FIG. 1A provides a flowchart of a process of air conditioning, in accordance with an exemplary embodiment of the present invention.

[0029] FIG. 1B provides a flowchart of a process of air conditioning, in accordance with an exemplary embodiment of the present invention.

[0030] FIG. 2 provides a flowchart of a process of air conditioning, in accordance with an exemplary embodiment of the present invention.

[0031] FIG. 3 provides a schematic of an air conditioning system, in accordance with an exemplary embodiment of the present invention.

[0032] FIG. 4 provides a schematic of an air conditioning system, in accordance with an exemplary embodiment of the present invention.

[0033] FIG. 5 provides a transition diagram of a polymer, in accordance with an exemplary embodiment of the present invention.

[0034] FIG. 6 provides a block diagram of a dehumidifier, in accordance with an exemplary embodiment of the present invention.

[0035] FIG. 7 provides a schematic of an air conditioning system, in accordance with an exemplary embodiment of the present invention.

[0036] FIG. 8 provides a schematic of an air conditioning system, in accordance with an exemplary embodiment of the present invention.

[0037] FIG. 9 provides an example dessicant wheel workflow according to the prior art.

[0038] FIG. 10A provides a psychrometric chart, in accordance with an exemplary embodiment of the present invention.

[0039] FIG. 10B provides a psychrometric chart, in accordance with an exemplary embodiment of the present invention.

[0040] FIG. 10C provides a psychrometric chart, in accordance with an exemplary embodiment of the present invention.

[0041] FIG. 11 provides a schematic of an air conditioning system, in accordance with an exemplary embodiment of the present invention.

[0042] FIG. 12 provides a schematic of an air conditioning system, in accordance with an exemplary embodiment of the present invention.

[0043] FIG. 13 provides images overlayed on a plot showing swelling of a polymer by temperature, in accordance with an exemplary embodiment of the present invention.

[0044] FIG. 14 provides a chart descriptive of the behavior of the cycle described herein, in accordance with an exemplary embodiment of the present invention.

[0045] FIG. 15 provides a chart descriptive of the behavior of the cycle described herein, in accordance with an exemplary embodiment of the present invention.

[0046] FIG. 16 provides a chart descriptive of the behavior of the cycle described herein and the traditional desiccant cycle, in accordance with an exemplary embodiment of the present invention.

DETAILED DESCRIPTION

[0047] To facilitate an understanding of the principles and features of the present disclosure, various illustrative embodiments are explained below. The components, steps, and materials described hereinafter as making up various elements of the embodiments disclosed herein are intended to be illustrative and not restrictive. Many suitable components, steps, and materials that would perform the same or similar functions as the components, steps, and materials described herein are intended to be embraced within the scope of the disclosure. Such other components, steps, and materials not described herein can include, but are not limited to, similar components or steps that are developed after development of the embodiments disclosed herein.

[0048] FIG. 1A provides an exemplary air conditioning process 100. The air conditioning process 100 can include a latent cooling stage including absorbing 102, via a responsive hygroscopic material, moisture from ambient air, heating 104, via a heat source, the responsive hygroscopic material above a transition temperature wherein the responsive hygroscopic material transitions from being hygroscopic to being hydrophobic, expelling 106, from the responsive hygroscopic material, liquid, and sensible cooling 108 the responsive hygroscopic material below the transition temperature.

[0049] As shown in FIG. 1B, the air conditioning process 100 can further include a sensible cooling stage including

sensible cooling 110, via an evaporative cooler, the ambient air. The evaporative cooler evaporates the liquid expelled from the responsive hygroscopic material. The process can further include moving 112 the responsive hygroscopic material from a first position to a second position. Absorbing 102, via a responsive hygroscopic material, moisture from ambient air step can be performed at the first position. Heating 102, via the heat source, the responsive hygroscopic material above a transition temperature step can be performed at the second position.

[0050] In any of the embodiments disclosed herein, the evaporative cooler can be one of a direct evaporative cooler, indirect evaporative cooler, or a partially direct and partially indirect evaporative cooler.

[0051] In any of the embodiments disclosed herein, the latent and sensible cooling of the ambient air is performed without the use of conventional refrigerants.

[0052] As shown in FIG. 2, the process can further include transferring 114 at least a portion of the heat removed from the hygroscopic material by sensible cooling the responsive hygroscopic material below the transition temperature, storing 116 the heat in the recuperator, and transferring 118 at least a portion of the stored heat back to the responsive hygroscopic material.

[0053] FIG. 3 provides an air conditioning system 300 according to the present disclosure. The system 300 can include a latent cooling stage 310 including a responsive hygroscopic material 312 configured to absorb moisture 314a from ambient air, a heat source 316 configured for heating the responsive hygroscopic material 312. The heat source 316 can be configured to heat the responsive hygroscopic material 312 above a transition temperature wherein the responsive hygroscopic material 312 transitions from being hygroscopic to being hydrophobic causing the responsive hygroscopic material 312 to expel a liquid 314b previously absorbed from the ambient air. The system 300 can further include a heat exchanger 318 configured for sensible cooling the responsive hygroscopic material 312 below the transition temperature.

[0054] In any of the embodiments disclosed herein, the air conditioning system can further include a sensible cooling stage 320 including an evaporative cooler 322 configured for sensible cooling the ambient by evaporating the liquid 314b expelled from the responsive hygroscopic material 312. In other words, FIG. 3 shows the thermodynamic cycle that forms the basis of the air conditioning system.

[0055] In any of the embodiments disclosed herein, the evaporative cooler 322 can be a direct evaporative cooler, an indirect evaporative cooler, or a partially direct and partially indirect evaporative cooler.

[0056] In any of the embodiments disclosed herein, the latent and sensible cooling of the ambient air is performed without the use of conventional refrigerants.

[0057] In any of the embodiments disclosed herein, the air conditioning system 300 can include an actuator configured to move the responsive hygroscopic material 312 from a first position to a second position. At the first position, the responsive hygroscopic material 312 can absorb moisture 314a from the ambient air. At the second position, the heat source 316 can heat the responsive hygroscopic material 312 above the transition temperature. Alternatively, or in addition, the air conditioning system 300 can further include a first polymer bed 710 and a second polymer bed 720 each including the responsive hygroscopic material 312. The air

conditioning system **300** can include one or more actuators which can be configured to change the airflow, as between inside air and outside air, passing over each of the first polymer bed **710** and second polymer bed **720**. The one or more actuators can control heat flow to the first polymer bed **710** and the second polymer bed **720** by controlling which airflow passes over the first polymer bed **710** and the second polymer bed **720**. For example, the one or more actuators can direct heat to the polymer bed that needs regeneration and away from the polymer bed that is absorbing moisture.

[0058] In any of the embodiments disclosed herein, the air conditioning system **300** can further include a recuperator **350**. The recuperator **350** can be configured to transfer at least a portion of the heat removed from the hygroscopic material **312** by sensible cooling the responsive hygroscopic material **312** below the transition temperature, storing the heat in the recuperator **350**, and transferring at least a portion of the stored heat back to the responsive hygroscopic material **312**.

[0059] In any of the embodiments disclosed herein, the responsive hygroscopic material can include PNIPAAm poly(N-isopropylacrylamide) and one or more hydrogels.

[0060] In any of the embodiments disclosed herein, the heat source can include one or more of solar heat, waste heat, gas heat, or electric heat.

[0061] In any of the embodiments disclosed herein, the heat exchanger **318** can include an ambient heat exchanger.

[0062] Stated otherwise, the present disclosure provides an air conditioning system **300** including a latent cooling stage **310**. The latent cooling stage **310** can include a responsive hygroscopic material **312** configured to absorb moisture **314a** from ambient air, a heat source **316** configured to heat the responsive hygroscopic material **312** above a transition temperature wherein the responsive hygroscopic material **312** transitions from being hygroscopic to being hydrophobic causing the responsive hygroscopic material **312** to expel a liquid **314b** previously absorbed from the ambient air, and a heat exchanger **318** configured for sensible cooling the responsive hygroscopic material **312**. The present disclosure can further include a sensible cooling stage **320** which can include an evaporative cooler **322** configured for sensible cooling the ambient by evaporating the liquid **314b** expelled from the responsive hygroscopic material **312**.

[0063] In any of the embodiments disclosed herein, the air conditioning system **300** can include a recuperator **350** configured to transfer at least a portion of the heat removed from the hygroscopic material **312**. Transferring at least the portion of the heat removed from the hygroscopic material **312** can be done by sensible cooling the responsive hygroscopic material **312** below the transition temperature, storing the heat in the recuperator **350**, and transferring at least a portion of the stored heat back to the responsive hygroscopic material **312**.

[0064] FIG. 7 shows a schematic of a system operating on the thermodynamic cycle disclosed herein. FIG. 7 shows the optional evaporative cooling stage and is implemented when both the humidity and temperature of the air must be lowered. When the evaporative cooler is used, the liquid water expelled from the regenerating polymer bed can also optionally have its excess heat rejected to the ambient environment prior to being sent to the evaporative cooler (i.e. heat exchanger between the regenerating polymer bed

and the evaporative cooler). This option reduces the water consumption in the evaporative cooler.

[0065] As highlighted in FIG. 8, recuperation is achieved by the addition of a regenerative thermal storage tank **810**. The regenerating bed (dry) is sensibly cooled by the storage medium. Then the bed must be further cooled by the ambient heat exchanger. Then the regenerating bed (swelled with water) is sensibly heated by the storage medium. Then, the bed is further heated by the heat source.

[0066] The following examples further illustrate aspects of the present disclosure. However, they are in no way a limitation of the teachings or disclosure of the present disclosure as set forth herein.

EXAMPLES

[0067] Disclosed herein is an example of a new desiccant air conditioning cycle that uses thermoresponsive polymers instead of traditional desiccants. A combined first and second law analysis demonstrates that this new cycle has at least three major advantages relative to the traditional case: (i) it can regenerate at lower temperatures, (ii) it can harvest liquid water and (iii) it has significantly higher coefficient of performances (COPs). For example, this new cycle can achieve a COP of 5.4 when regenerated at 100° C., whereas the traditional desiccant cycle is limited to a COP of ~1. The fundamental origins of these advantages can be traced to the method of regeneration. The traditional desiccant cycle regenerates by flowing hot air over the desiccant, which provides a medium for gaseous water desorption. However, this also generates entropy and places a minimum temperature constraint on the hot air. In contrast, the thermoresponsive polymer cycle regenerates through a polymer phase transition. The polymer absorbs water vapor in humid air, and then expels liquid water when raised above its transition temperature. This regeneration method can be done via direct heating, generates liquid water, and relaxes constraints on entropy generation and minimum temperature. The minimum regeneration temperature of the thermoresponsive cycle is only limited by the transition temperature of the polymer, which can be tuned through materials science. Due to its liquid water harvesting capability, the new cycle potentially eliminates water consumption when used with evaporative cooling, or it can be directly used for atmospheric water harvesting.

[0068] The adoption of efficient and environmentally sustainable space cooling technologies is a significant challenge facing society. Vapor compression systems remain the most widely adopted method of air conditioning, but their refrigerants have high global warming potentials, and they are often inefficient at dehumidification. Desiccant cycles are an attractive alternative to vapor compression cycles. When paired with evaporative cooling, desiccant systems can independently control humidity and temperature, thereby improving occupant comfort without using refrigerants. Additionally, desiccant cycles are heat-driven, allowing them to take advantage of solar- or waste-heat to reduce electricity consumption. However, the adoption of desiccant cycles has been hindered by low coefficients of performance (COPs), high regeneration temperatures, and, when paired with evaporative cooling, consumption of water.

[0069] Thermoresponsive polymer gels have recently garnered attention for their ability to sorb water vapor from humid air and expel it as liquid. This behavior arises from the phase separation of thermoresponsive polymer solutions

at critical solution temperatures. Thermoresponsive polymers exhibit a lower critical solution temperature (LCST) or upper critical solution temperature (UCST). In the case of polymers with an LCST, a single-phase polymer solution forms below the LCST, and a two-phase mixture forms above the LCST. UCST behavior is the reverse, where separation occurs below the UCST and mixing occurs above the UCST. Of these two behaviors, this example focuses on LCST behavior. If an LCST polymer crosslinks, the resulting gel will have a transition temperature (T_t), below which swelling occurs and above which shrinking occurs. In general, the gel transition temperature is often near (or identical to) the polymer solution LCST. This swelling and shrinking behavior can occur in liquid water or humid air.

[0070] This example provides a new thermodynamic cycle that utilizes LCST polymer gels to dehumidify air and harvest liquid water. To analyze this new cycle, the inventors apply mass balances to the air, polymer, and water, along with the first and second laws of thermodynamics to relate the heat flow and entropy generation in each component of the cycle. The entropy generation is set to zero to find the minimum enthalpy and entropy of polymer swelling that are required to produce the desired dehumidification for a given polymer transition temperature. The inventors then use the minimum enthalpy of swelling to find the limit to the cycle efficiency.

[0071] The inventors also utilize the first and second laws to analyze a traditional desiccant cycle and compare it to the new thermoresponsive polymer cycle. The inventors show that the traditional desiccant cycle has a lower limit to the regeneration temperature, which does not exist for the thermoresponsive polymer desiccant cycle. Realistically, the traditional desiccant cycle must be regenerated with temperatures higher than the lower limit. Higher regeneration temperatures introduce entropy generation in the traditional desiccant cycle and limit the COP of these cycles to approximately 1. Conversely, the thermoresponsive polymer cycle can be regenerated without entropy generation at higher temperatures, meaning the COP of this new cycle is not limited to a value of 1 and can be significantly higher than the traditional desiccant COP.

[0072] Herein there is provided a heat-driven thermoresponsive polymer desiccant cooling and dehumidification cycle that utilizes an LCST polymer gel as the desiccant (LCST cycle).

[0073] The behavior of LCST materials can be explained through the free energy of swelling and shrinking according to the equation $g_{shrink} = h_{shrink} - TS_{shrink} = -g_{swell}$.

[0074] Below the transition temperature of an LCST gel, g_{swell} is negative and the gel swells. Above the transition temperature, g_{shrink} is negative and the polymer shrinks, thereby expelling the water. At the transition temperature, the free energy of swelling and shrinking are both zero, giving the condition $h_{shrink,t} = T_t s_{shrink,t}$. The subscript “t” refers to the enthalpy and entropy of shrinking evaluated at the transition temperature, as these quantities generally vary with temperature. The temperature variation of h_{shrink} and s_{shrink} is due to a change in specific heat that accompanies swelling/shrinking (cp).

[0075] In general, humid, ambient air enters the LCST cycle at the inlet and is separated into dry air and liquid water (also at ambient temperature) at the outlets. Some heat is rejected to the ambient (Q_H), and some heat must be delivered to the cycle (QS) from a heat source with a

temperature greater than ambient. To achieve this overall cycle, four main processes must occur, illustrated in FIG. 3: swelling of the gel in humid air, sensible heating of the gel, shrinking of the gel to expel liquid water, and sensible cooling back to the original temperature. The swelling process serves to dehumidify the air, while the other three processes (sensible heating, shrinking, and cooling) serve to regenerate the gel by removing the water. Aside from dehumidification, the LCST cycle can also be used for cooling. To use the cycle for cooling, the outlet liquid water is evaporated in the dry outlet air, thereby lowering the air temperature.

[0076] The T-s diagram of a polymer undergoing the LCST cycle (FIG. 5) is analogous to the Ericsson cycle. The entropy of the polymer first decreases with a magnitude of $s_{swell,amb}$ as it absorbs water vapor at T_{amb} . Then, when the polymer is sensibly heated, the entropy and temperature both increase. Next, as the polymer undergoes latent heating at T_t , the entropy increases by $s_{shrink,t}$ as liquid water is expelled. Finally, the polymer is sensibly cooled, and entropy and temperature decrease. The shape of the LCST cycle T-s diagram in FIG. 5 is the same as the Ericsson cycle, and similarities between the two cycles exist. Just as the Ericsson cycle approaches the Carnot efficiency when recuperation is used, the LCST cycle approaches the COP of a heat-driven Carnot refrigerator when recuperation is used, as shown herein. To evaluate the upper limits of the LCST cycle, the inventors analyze a continuous configuration of the cycle that contains a recuperator.

[0077] While FIG. 6 represents the cyclic or sequential operation of the LCST cycle, FIG. 4 illustrates the continuous mode of operation. During continuous operation, the gel physically moves through different control volumes. Here, a recuperator can be added to decrease the required sensible heat input, which allows the thermodynamic limits of the cycle to be analyzed. While this recuperation process may be difficult to implement in practice, the cycle can be analyzed without recuperation by setting the recuperator effectiveness, f , to zero.

[0078] One hygroscopic material known in the art is an interpenetrating polymer network with poly(N-isopropylacrylamide) (PNIPAm) and sodium alginate. The PNIPAm network provides the thermoresponsive behavior, while the sodium alginate increases hygroscopicity. The gels absorb water from the atmosphere when below the transition temperature and expel liquid water when the temperature is raised. Another is a gel with a PNIPAm network and a polypyrrole chloride network. These gels swell with a significant amount of water per unit mass of dry polymer.

[0079] The LCST cycle requires polymers with higher enthalpies of shrinking, and thus more heat input, when the polymer sorbs water from low humidity air. Consequently, the LCST cycle will be less efficient when used in dry areas, like those where water is scarce and atmospheric water harvesting might be attractive. Conversely, the LCST cycle will be more efficient in highly humid regions, where air conditioning use is highly desirable.

[0080] Relating to the mass balance of the LCST cycle—the LCST cycle’s mass balance is achieved by relating the air mass flow rate (m_a), liquid water mass flow rate (m_{H_2O}), polymer mass flow rate (m_p), inlet and outlet humidity ratios (w_1 and w_2 , respectively), and the polymer swelling ratio (C , the ratio of water uptake to dry polymer mass). The resulting mass balance equation is:

$$mp = \frac{m_{H_2O}}{\Delta C} = \frac{m_a(w_1 - w_2)}{\Delta C}$$

[0081] Relating to first and second law analysis of the LCST cycle The first and second laws of thermodynamics are used to find the heat inputs required to drive the LCST cycle.

[0082] From FIG. 4, there are sensible and latent heat inputs. The sensible heat is required to bring the swelled polymer and absorbed liquid water from the temperature they possess as they leave the recuperator, $T_{int,1}$, to the transition temperature, T_t . For all the cases analyzed in this paper, the specific heat increment of polymer swelling is assumed to be zero ($\Delta c_p=0$), but provided herein are results for a gel with a non-zero Δc_p . When $\Delta c_p=0$, $T_{int,1}$ is related to the recuperator effectiveness through the expression given in Equation 1. When the recuperator effectiveness is unity, no sensible heating is required.

$$T_{int,1} = T_{amb} + \epsilon(T_t - T_{amb}) \quad \text{Equation 1}$$

[0083] The second heat input to the LCST cycle is the latent heat, or the enthalpy of shrinking of the thermoresponsive polymer, $\Delta h_{shrink,t}$. However, because this thermodynamic cycle must obey the Carnot limit, there must be a limit to the value of $\Delta h_{shrink,t}$. FIG. 6 shows the dehumidifier control volume, with the enthalpies and entropies at the inlet and outlet states. Applying the first and second laws of

entropies, the enthalpy and entropy of liquid water (h_f and s_f respectively), the ambient temperature, the polymer transition temperature, the change in polymer specific heat that accompanies the shrinking transition (Δc_p), and the polymer swelling ratio. The enthalpies h_1 and h_2 and entropies s_1 and s_2 are evaluated for humid air (i.e. a mixture of dry air and water vapor) and are fixed by the temperature and humidities at states **1** and **2**.

$$\Delta h_{shrink,t,rev} = \frac{\Delta C}{\frac{1}{T_{amb}} - \frac{1}{T_t}} \left(\frac{s_1 - s_2 - \left(\frac{h_1 - h_2}{T_{amp}} \right)}{w_1} - \left(s_f - \frac{h_f}{T_{amb}} \right) \right) + \frac{\Delta c_p}{\frac{1}{T_{amb}} - \frac{1}{T_t}} \frac{\ln\left(\frac{T_{amb}}{T_t}\right) - (T_{amb} - T_t)}{T_{amb}} \quad \text{Equation 3}$$

[0085] The enthalpy of shrinking is expressed in Equation 3 in units of energy per unit mass of dry polymer. Alternatively, the enthalpy of shrinking can be expressed per unit mass of water absorbed by the polymer, given in Equation 4. When expressed in this way, enthalpy of shrinking is referred to as $\Delta h_{shrink,t,rev,H_2O}$.

$$\Delta h_{shrink,t,rev,H_2O} = \frac{\left(\frac{s_1 - s_2 - \left(\frac{h_1 - h_2}{T_{amp}} \right)}{w_1 - w_2} - \left(s_f - \frac{h_f}{T_{amb}} \right) + \frac{\Delta c_p}{\Delta C} - \frac{\ln\left(\frac{T_{amb}}{T_t}\right) - (T_{amb} - T_t)}{T_{amb}} \right)}{\frac{1}{T_{amb}} - \frac{1}{T_t}} \quad \text{Equation 4}$$

thermodynamics to this control volume, as well as the previously mentioned relationship between the enthalpy and entropy of swelling, yields Equation 2.

$$\frac{s_{gen,dehum}}{m_a} = \frac{(h_1 - h_2)}{T_{amb}} - (s_1 - s_2) + (w_1 - w_2) \left(\frac{h_f}{T_{amb}} - s_f \right) + \frac{w_1 - w_2}{\Delta C} \Delta h_{swell,t} \left(\frac{1}{T_t} - \frac{1}{T_{amb}} \right) + \frac{w_1 - w_2}{\Delta C} \Delta c_p \left(\frac{T_{amb} - T_t}{T_{amb}} + \ln\left(\frac{T_{amb}}{T_t}\right) \right) \quad \text{Equation 2}$$

[0084] Setting the entropy generation in the dehumidifier to zero ($S_{gen,dehum}=0$) yields the reversible limit for the enthalpy of swelling. Recognizing that the enthalpy of swelling and enthalpy of shrinking are equal in magnitude and opposite in sign, the reversible enthalpy of shrinking, $\Delta h_{shrink,t,rev}$, is derived, shown in Equation 3. This is the minimum enthalpy of shrinking that an LCST gel would need to bring air from the inlet humidity of w_1 to the outlet humidity of w_2 , when sorption occurs at T_{amb} and the polymer has a transition temperature of T_t . This reversible enthalpy of shrinking is a function of the inlet air (state **1**) and outlet air (state **2**) humidity ratios, enthalpies, and

[0086] The cooling power of the cycle, Q_C , depends on its intended use. When used as a dehumidifier, the LCST cycle's useful effect is the removal of water vapor from the $Q_{C,dehum} = m_{H_2O} h_g = m_a(w_1 - w_2) h_g$, where h_g is the enthalpy of water vapor. In this use case, the outlet liquid water stream is discarded. When the system is used for cooling, the liquid water is adiabatically recombined with the air to lower its temperature (through evaporative cooling). In this case, the useful cooling effect is $Q_{C,cool} = m_{H_2O} h_{fg} = m_a(w_1 - w_2) h_{fg}$, since the liquid water enthalpy is being added back to the air. The heat supplied to the system is the sum of the sensible heat to raise the swelled polymer temperature and the latent heat (enthalpy of shrinking): The COP of the cycle (COP) is defined as $COP = Q_C / Q_S$, where Q_C is different for the dehumidification and cooling use cases as described above. The COP takes the form of Equation 5 and Equation 6 when the cycle is being used as a dehumidifier and cooler, respectively.

$$COP_{dehum} = \frac{\Delta C * h_g}{(c_{p,swell} + \Delta C * c_{p,H_2O})(T_t - T_{int,1}) + \Delta h_{shrink}} \quad \text{Equation 5}$$

$$COP_{cool} = \frac{\Delta C * h_{fg}}{(c_{p,swell} + \Delta C * c_{p,H_2O})(T_t - T_{int,1}) + \Delta h_{shrink}} \quad \text{Equation 6}$$

[0087] The COP can also be calculated using effective temperatures of heat transfer and the “second law” COP in Equation 7. The first term on the right-hand side of Equation 7 is the Carnot COP of a heat-driven cooling cycle, while the second term accounts for internal irreversibility. The temperatures in Equation 7 are effective temperatures, derived from a second law analysis of the cycle, and must be expressed in absolute units of temperature. T_S is the effective temperature of heat addition, T_C is the effective temperature of dehumidification/cooling, and T_H is the effective temperature of heat and mass rejection to ambient. These effective temperatures can be used to identify entropy generation external to the cycle, while S_{gen} is the internal entropy generation. Equation 7 can be used to find either COP_{dehum} or COP_{cool} , where the equations for T_H and T_C are different depending on whether dehumidification or cooling is being considered.

$$COP = \left(1 - \frac{T_H}{T_S}\right) \frac{T_C}{T_H - T_C} - \frac{\dot{S}_{gen}}{\dot{Q}_S} \frac{T_H T_C}{T_H - T_C} \quad \text{Equation 7}$$

[0088] The effective temperatures are derived by applying the first and second laws to the cycle. The expression for the effective temperature of heat addition, T_S , is given in Equation 8.

$$T_S = \frac{(c_{p,swell} + \Delta C * c_{p,H2O})(T_t - T_{int,1} + \Delta h_{shrink,t})}{(c_{p,swell} + \Delta C * c_{p,H2O}) \ln\left(\frac{T_t}{T_{int,1}}\right) + \Delta s_{shrink,t}} \quad \text{Equation 8}$$

[0089] The effective temperature of dehumidification/cooling, T_C , and the effective temperature of heat and mass rejection to the ambient, T_H , depend slightly on the intended use of the cycle. When the cycle is used for dehumidification, the expression for T_C takes the form in Equation 9. When the cycle is used for cooling, T_C is described by Equation 10.

$$T_{C,dehum} = \frac{h_1 - h_2}{s_1 - s_2} \quad \text{Equation 9}$$

$$T_{C,cool} = \frac{h_1 - h_2 - (w_1 - w_2)h_f}{s_1 - s_2 - (w_1 - w_2)s_f} \quad \text{Equation 10}$$

[0090] Equation 11 gives the expression for T_H when the cycle is used for dehumidification, and Equation 12 describes T_H when the cycle is used for cooling.

$$T_{H,dehum} = \left(h_2 - h_1 + \frac{w_1 - w_2}{\Delta C} \Delta h_{swell,amb} + \frac{w_1 - w_2}{\Delta C} (c_{p,shrink} + \Delta C * c_{p,H2O}) * (T_{amb} - T_{int,2})\right) * \left(s_2 - s_1 + \frac{w_1 - w_2}{\Delta C} \Delta s_{swell,amb} + \frac{w_1 - w_2}{\Delta C} (c_{p,shrink} + \Delta C * c_{p,H2O}) * \ln\left(\frac{T_{amp}}{T_{int,2}}\right)\right)^{-1} \quad \text{Equation 11}$$

-continued

Equation 12

$$T_{H,cool} = \left(h_2 + (w_1 - w_2)h_f - h_1 + \frac{w_1 - w_2}{\Delta C} \Delta h_{swell,amb} + \frac{w_1 - w_2}{\Delta C} (c_{p,shrink} + \Delta C * c_{p,H2O}) * (T_{amb} - T_{int,2})\right) * \left(s_2 + (w_1 - w_2)s_f - s_1 + \frac{w_1 - w_2}{\Delta C} \Delta s_{swell,amb} + \frac{w_1 - w_2}{\Delta C} (c_{p,shrink} + \Delta C * c_{p,H2O}) * \ln\left(\frac{T_{amp}}{T_{int,2}}\right)\right)^{-1}$$

[0091] When the system is internally reversible, Equation 7 reduces to the Carnot COP of a heat-driven cycle: $COP = (1 - T_H/T_S) T_H T_C / (T_H - T_C)$.

[0092] If the recuperator has an effectiveness less than unity, the need to sensibly heat the gel makes the cycle both externally and internally irreversible. External irreversibility increases the value of T_H and reduces the value of T_S . When the system is externally reversible, T_H and T_S reach their ideal values. For example, when the LCST cycle is externally reversible and used for cooling, $T_H = T_{amb}$ and $T_S = T_r$. Even though Equation 7 yields the same value as Equation 5 or 6 (depending on the equations used for T_C and T_H), it provides important insight. Specifically, the rate of entropy generation and the effective temperatures are useful for identifying points of internal and external irreversibility and their effect on the cycle COP.

[0093] A simple configuration of the traditional desiccant cycle is given in FIG. 10A. Humid, ambient air enters a rotating wheel filled with solid desiccant. The air enters with ambient conditions at state 1 and leaves hot and dry at state 1a. The hot and dry air at state 1a is sent through a heat exchanger, which has humid regeneration air at state 3 entering on the other side. The hot and dry supply air is cooled to the ambient temperature by the heat exchanger, which results in dry air at ambient temperature leaving the system (just as in the LCST cycle) at state 2. On the other side of the heat exchanger, the humid regeneration air is preheated to state 4. We note that our choice of state numbers is deliberately chosen such that states 1 and 2 in the traditional desiccant cycle are equivalent to states 1 and 2 in the LCST cycle, and hence our inclusion of the curiously-named state 1a in FIG. 4.

[0094] FIG. 9 shows a traditional desiccant cycle schematic and FIGS. 10A-10C show psychrometric charts. The psychrometric chart in FIG. 10A shows the scenario where the regeneration temperature (T_6) is at its minimum possible value ($T_{6,min}$). This occurs when the relative humidity at state 6 is equal to the relative humidity at state 1a. In this case, T_7 is lower than T_4 , so the recuperator is not needed and is bypassed. The chart in FIG. 10B shows the scenario where T_6 is equal to $T_{6,crit}$, which occurs when T_7 is equal to T_4 , and the recuperator provides no use. The psychrometric chart in FIG. 10C shows the scenario where T_6 is greater than $T_{6,crit}$ and T_7 is greater than T_4 . In this case, the recuperator is useful, as uses the air leaving the desiccant at state 7 to preheat the regeneration air to state 5.

[0095] A desiccant is able to dehumidify air because the chemical potential of the water vapor in the air is equal to (reversible) or greater than (realistic/irreversible) the chemical potential of the water in the desiccant. For a desiccant to dehumidify air to state 1a, the desiccant must first be dried with air that has an equal or lower chemical potential than state 1a. In other words, state 6 in FIG. 4 must have an equal

or lower chemical potential than state **1a**. For small differences in temperature between state **1a** and state **6**, this is approximately equivalent to state **6** having an equal or lower relative humidity than state **1a**. Thus, the minimum regeneration temperature ($T_{6,min}$), is the one that results in state **6** having the same relative humidity as state **1a**; this is pictured in FIG. **10A**. Realistic regeneration temperatures would be higher and would result in state **6** having a lower relative humidity than state **1a**, as pictured in FIG. **10B-10C**.

[0096] For the case pictured in FIG. **10A**, T_7 is lower than T_4 , so the regeneration air leaving the wheel cannot be used to preheat the regeneration air at state **4**. In this case, the recuperator would be bypassed. In FIG. **10B**, $T_6=T_{6,crit}$, which occurs when $T_7=T_4$, so no heat transfer would occur in the recuperator. When $T_6>T_{6,crit}$, as pictured in FIG. **10C**, $T_7>T_4$, so the recuperator is used to further preheat the regeneration air from state **4** to state **5**. Thus, the traditional desiccant is analyzed without the recuperator for $T_{6,min}<T_6<T_{6,crit}$ and with the recuperator for $T_6>T_{6,crit}$. For the traditional desiccant cycle in FIG. **9**, a value of 1 is used for the heat exchanger effectiveness and recuperator effectiveness (when the recuperator is used).

[0097] The effective temperatures of heat transfer can be derived for the traditional desiccant cycle in the standard approach. Heat addition occurs from states **5** to **6**, where the effective temperature $T_{S,des}$ is given in Equation 13. The useful effect of the cycle, dehumidification, occurs between states **1** and **2**, resulting in Equation 14 for $T_{C,des}$. Regeneration air enters the cycle at state **3** and is exhausted to ambient at state **8**; Equation 15 gives the expression for $T_{H,des}$. These effective temperatures are for dehumidification, not cooling, as the traditional desiccant cycle does not harvest liquid water for evaporative cooling.

$$T_{S,des} = \frac{h_6 - h_5}{s_6 - s_5} \quad \text{Equation 13}$$

$$T_{C,des} = \frac{h_2 - h_1}{s_2 - s_1} \quad \text{Equation 14}$$

$$T_{H,des} = \frac{h_8 - h_3}{s_8 - s_3} \quad \text{Equation 15}$$

[0098] FIG. **14** shows Reversible (or minimum possible) enthalpy of shrinking for varying relative humidities (RH) at an ambient temperature of 30° C. and a polymer swelling ratio of 0.1 kg/kg.

[0099] FIG. **14** shows that absorbing water vapor requires higher enthalpies of shrinking in dry air than in humid air. All curves in FIG. **14** correspond to the same decrease in relative humidity (2% between the inlet and outlet), but the relative humidity at the inlet is different for each curve. As the inlet relative humidity decreases for a given transition temperature, higher enthalpies of shrinking are required to achieve the same decrease in relative humidity from inlet to outlet.

[0100] FIG. **14** also shows that the minimum possible enthalpy of shrinking increases rapidly as the transition temperature (T_t) approaches the temperature at which swelling occurs (T_{amb}); as the difference between T_t and T_{amb} decreases, so does the polymer's affinity for swelling. Thus, Δh_{swell} must be very large in magnitude to compensate and drive the swelling process when T_t is near T_{amb} .

[0101] FIG. **16** shows that in the reversible limit, the LCST cycle COP increases monotonically as a function of transition temperature. To calculate the COP, the inventors used the reversible enthalpy of shrinking from Equation 3 and set $\Delta c_p=0$. When $\Delta c_p=0$ and $f=1$, the recuperator is reversible, and no entropy is generated; thus, the $f=1$ curve in FIG. **16** represents the Carnot COP. At low transition temperatures, the COP is low, because the enthalpy of shrinking must be large in magnitude to drive dehumidification.

[0102] The COP peaks at a finite transition temperature for recuperator effectiveness values less than unity (FIG. **15**). Below the peak temperature, the COP is limited by a large $\Delta h_{shrink,t}$ value, while above the peak temperature the COP is limited by sensible heating.

[0103] FIG. **15**. LCST cycle COP_{cool} as a function of transition temperature, with different curves for various recuperator effectiveness values and swelling ratios. For all curves, ambient temperature is 30° C., inlet relative humidity is 66%, and outlet relative humidity is 40%.

[0104] The COP increases with increasing swelling ratios when the recuperator effectiveness is less than unity (dotted lines in FIG. **15**). A polymer with a higher swelling ratio would require less polymer mass to remove the same amount of water from the air. Less polymer mass reduces the sensible heating, which increases the system COP. The solid lines in FIG. **15** illustrate that an increased recuperator effectiveness leads to an increased COP. FIG. **15** also shows that a polymer with a swelling ratio of 10 kg/kg and no recuperator would have a COP nearly equal to the COP of a polymer with a swelling ratio of 0.1 kg/kg and a recuperator effectiveness of 0.9. When the swelling ratio is high, the recuperator can be eliminated without significantly affecting the COP.

[0105] The COPs of the LCST cycle and traditional desiccant cycle are plotted as functions of regeneration temperature in FIG. **16**. The COPs of the two cycles are very close when both cycles have a regeneration temperature equal to $T_{6,min}$ (the minimum possible regeneration temperature for the traditional desiccant cycle). FIG. **16** highlights two advantages of the LCST cycle. Firstly, the LCST cycle can operate with lower regeneration temperatures than the traditional desiccant cycle. The LCST of PNIPAm, a common thermoresponsive polymer, is 32° C., and FIG. **16** shows that the LCST cycle could operate with such a polymer. The COP would be low for such a polymer, but the regeneration temperature is low enough that almost any common heat source could be used to regenerate the polymer, including "free" heat sources like solar- and waste-heat. On the other hand, the traditional desiccant cycle could not operate with a regeneration temperature of 32° C., as the minimum regeneration temperature ($T_{6,min}$) is greater than 32° C. In other words, if the traditional desiccant were regenerated with a 32° C. heat source, it would not be able to dehumidify the air at state **2** to 40% relative humidity, but the LCST cycle could.

[0106] FIG. **16**. Shows the LCST cycle and traditional desiccant COP_{dehum} as a function of regeneration temperature. For all curves, ambient temperature is 30° C., inlet relative humidity is 66%, and outlet relative humidity is 40%. For both cycles, the recuperator effectiveness values are 1. The regeneration temperature on the x-axis is T_t for the LCST cycle and T_6 for the traditional desiccant cycle.

[0107] The second major advantage of the LCST cycle is that the LCST cycle COP increases with regeneration temperature, while the traditional desiccant cycle COP decreases. To improve the cycle COP, an LCST gel with a higher transition temperature could be synthesized and utilized in the LCST cycle. At these higher regeneration temperatures, it is clear that the LCST cycle could achieve COPs several times higher than traditional desiccant cycles. For example, the LCST cycle has a maximum COP of 5.4 when the polymer transition temperature is 100° C., compared to the traditional desiccant cycle maximum COP of 1 at the same regeneration temperature. When the recuperator effectiveness is unity, the LCST cycle COP monotonically increases with the regeneration temperature (T_r), approaching an asymptote far greater than 1. The existence of this asymptote can be seen in the functional form of Equation 7, and it occurs at temperatures well over 1000° C. for the conditions in FIG. 16. Temperatures this high are unlikely to be used in the LCST cycle, so we limit the FIG. 16 results to typical desiccant regeneration temperatures. In contrast to the LCST cycle COP, the traditional desiccant cycle COP monotonically decreases with regeneration temperature (T_6), approaching an asymptote of 1. While FIG. 16 shows that the traditional desiccant cycle would be fairly efficient if regenerated at $T_{6,min}$, this is unrealistic, as most traditional desiccants must be regenerated at temperatures closer to 100° C., depending on the operating conditions. When the regeneration temperature is above $T_{6,crit}$, the traditional desiccant cycle COP approximately reaches the limit of 1. This limitation is illustrated in FIG. 10C; the cooling effect (represented by the length between points 1a and 2) is equal to the heat supplied (represented by the length between points 5 and 6). Thus, traditional desiccants have been historically limited to COPs near 1.

the LCST cycle is important. For example, the DEVap system uses evaporative cooling and thus consumes water. Regeneration of traditional desiccants involves the exhausting of water as vapor to outside air. The LCST cycle expels liquid water during regeneration which can be harvested and reused with evaporative cooling. This presents the potential for a dehumidification and cooling cycle with net-zero water consumption (in sufficiently humid environments), which is something that traditional desiccant cycles cannot achieve.

[0109] In addition to evaporative cooling, the liquid water given off during regeneration of the LCST cycle can be used for atmospheric water harvesting.

[0110] Table 1 shows the differences between the traditional desiccant cycle and the LCST cycles. To make the cycles comparable, all were given recuperator effectiveness values of unity, and the inlet air (state 1) and outlet air (state 2) conditions were the same.

[0111] Table 1 shows effective temperatures of heat transfer, rate of entropy generation, and COP_{dehum} of two traditional desiccant cycle scenarios and two LCST cycle scenarios. For all four cases, the ambient temperature is 30° C., the inlet relative humidity is 66%, the outlet relative humidity is 40%, the mass flow rate of air is 1 kg/s, and $\epsilon=1$. Effective temperatures are kept in Kelvin, as they do not represent measurable temperatures; rather, they are equivalent temperatures derived from the second law that must be in absolute units when used to find the COP in Equation 7. The minimum regeneration temperature for the traditional desiccant, $T_{6,min}$, is 55.3° C., so the $T_{regen}=32$ ° C. is blank for the traditional desiccant, as it is thermodynamically impossible. The critical regeneration temperature for the traditional desiccant ($T_{6,crit}$), which is the temperature above which the recuperator becomes useful, is 63.6° C.

TABLE 1

	$T_C(K)$	$T_H(K)$	$T_S(K)$	$\dot{S}_{gen}(W/K)$	COP_{dehum}
LCST Cycle ($T_r = 32.0$ ° C.)	292.2	303.0	305.2	0	0.19
LCST Cycle ($T_r = 55.3$ ° C.)	292.2	302.6	328.4	0	2.21
LCST Cycle ($T_r = 63.6$ ° C.)	292.2	302.6	336.8	0	2.87
LCST Cycle ($T_r = 100.0$ ° C.)	292.2	302.5	373.2	0	5.40
Traditional Desiccant ($T_{regen} = 32.0$ ° C.)	—	—	—	—	—
Traditional Desiccant ($T_{regen} = 55.3$ ° C.)	292.2	301.2	323.8	0.2	2.02
Traditional Desiccant ($T_{regen} = 63.6$ ° C.)	292.2	304.5	327.9	4.2	1.03
Traditional Desiccant ($T_{regen} = 100.0$ ° C.)	292.2	304.7	364.9	7.2	1.01

[0108] It is worth noting that a third major advantage of the LCST cycle, not highlighted in FIG. 16, is that the LCST cycle harvests liquid water, which can be used to create an evaporative cooler with net-zero water consumption. This is an important advantage that sets the LCST cycle apart from traditional desiccant cycles, even in situations where the first two advantages of the LCST cycle do not prove to be important. For example, the DEVap system created at the National Renewable Energy Laboratory (NREL) is a liquid desiccant system that can achieve a COP of 1.2-1.4 and can be regenerated with solar- or waste-heat. While the LCST cycle could reach a COP higher than the DEVap system, that becomes less important when considering solar- and waste-heat are “free”. In that case, a higher COP (and thus a lower rate of heat input) becomes less impactful. However, even if the possibility of lower regeneration temperatures and higher COPs is not important, the liquid water harvesting of

[0112] All cycles in Table 1 have the same value for T_C , as they provide the same amount of dehumidification. The difference arises in the effective temperatures of the heat source and heat rejection, as well as the entropy generation.

[0113] Entropy is generated because of the mismatch in relative humidity between the supply and regeneration air streams that occurs at higher regeneration temperatures. Thus, the traditional desiccant cycle is limited by a minimum regeneration temperature, but it is inefficient at higher regeneration temperatures. There is, then, only one regeneration temperature for which the traditional desiccant cycle could reach the Carnot limit.

[0114] Conversely, the LCST cycle does not have a minimum regeneration temperature based purely on the properties of the air. This is because the LCST cycle regenerates by rejecting liquid water as opposed to gaseous water into the air. The LCST cycle regeneration temperature is the polymer

transition temperature, which only needs to be greater than ambient temperature. A polymer with a lower transition temperature than the minimum desiccant regeneration temperature could be more easily regenerated by solar- or waste-heat. Even though the heat input would be high due to the low transition temperature, the heat would come at no cost for using the cycle. On the other hand, if a high temperature heat source is available, a polymer with a high transition temperature could be used to create a more efficient system than the traditional desiccant. The traditional desiccant necessarily generates entropy at high regeneration temperatures, while the LCST cycle does not. A polymer with a high transition temperature could be synthesized to create a system that requires little heat input. Even the water harvesting of the LCST cycle is more convenient than the traditional desiccant cycle. For a traditional desiccant cycle to be used for cooling, an external water source would be required for evaporative cooling. In contrast, the LCST cycle harvests its own liquid water from the air that it dehumidifies.

[0115] The LCST cycle results in Table 1 were calculated using the reversible enthalpy of shrinking from Equation 3, which corresponds to zero entropy generation in the dehumidifier. Since the results were calculated for a recuperator effectiveness of unity, no entropy is generated in the recuperator. This means $S_{gen}=0$ for the LCST cycle, regardless of the regeneration temperature.

[0116] The highest temperature in the cycle is T_r , so the actual temperature of the heat source must be at least T_r . When $T_s < T_r$, not all of the exergy from the heat source is being utilized, and exergy is being destroyed (i.e. entropy is being produced) external to the cycle. When the recuperator in the LCST cycle has an effectiveness less than unity, the heat source (at a temperature of T_r) is used to increase the temperature of the swelled polymer to T_r . This means heat is being transferred between two bodies at different temperatures, and entropy is being externally generated, which reduces the value of T_s . The results in Table 1 were calculated for a recuperator effectiveness of 1, meaning no sensible heating is required. The lack of sensible heating maximizes T_s ; in fact, $T_s=T_r$ for all of the LCST cycle results in Table 1. This means that the effective and actual heat source temperatures are equal, and all of the exergy from the heat source is being used.

[0117] The ambient temperature for the results presented in Table 1 is 30° C., or 303.15 K; however, the effective temperature T_H is lower than 303.15 K for all of the LCST cycle results. This may seem impossible at first: heat cannot be rejected to ambient at a temperature lower than the ambient temperature. However, it is important to note that the effective temperature T_H also includes mass (namely, liquid water) that is exhausted to ambient, and that reduces the value below the actual ambient temperature. If Table 1 were recreated for COP_{cool} , all of the T_H values would be equal to 303.15 K for the LCST cycle. This is because the cooling configuration of the cycle considers the liquid water as part of the useful cooling effect and not as being rejected to ambient.

[0118] It should be noted that the liquid water exhausted to ambient actually has some exergy (as it does not have the same chemical potential as the 66% relative humidity ambient air). Thus, when used for dehumidification, the LCST cycle described in FIG. 4 is externally irreversible and is not 100% exergy efficient. A 100% efficient version of the

dehumidification configuration of the LCST cycle would have some way of scavenging the exergy from the exhausted liquid water, but that is beyond the scope of this paper. If the exergy of the liquid water were used instead of lost to ambient, the T_H values would be lower than those in Table 1.

[0119] While there is some small difference in T_H when comparing the LCST and traditional desiccant cycles, T_s and S_{gen} are the major factors that explain the deviation in performance between the LCST and traditional desiccant cycles. For the same regeneration temperature, the T_s value is lower for the traditional desiccant than for the LCST cycle. Thus, the traditional desiccant cycle is not able to use all of the exergy from the heat source that the LCST cycle is able to use.

[0120] While the traditional desiccant T_s values are lower than those of the LCST cycle, they still increase with increasing regeneration temperature. Despite the increasing T_s values, it is clear from Table 1 that the COP decreases as regeneration temperature increases. This might seem counterintuitive; a higher temperature heat source tends to increase the COP of a heat-driven cooling cycle, as seen in Equation 7. However, it is also clear from Equation 7 that internal entropy generation decreases the COP, and Table 1 shows that entropy generation in the traditional desiccant cycle increases with increasing regeneration temperature. This entropy generation is why the traditional desiccant cycle COP decreases as T_{regen} increases, while the lack of entropy generation in the LCST cycle is why its COP increases with T_{regen} . The reason why entropy generation increases with regeneration temperature in the traditional desiccant cycle is due to the method of regeneration.

[0121] The conditions of the regeneration air, namely the relative humidity or chemical potential, determine the amount of entropy generation in the traditional desiccant cycle. The ideal case is when the desiccant dehumidifies the process air to the same relative humidity (RH) as the regeneration air entering the desiccant (i.e. $R_{H1a}=R_{H6}$). This condition is described by $T_6=T_{6,min}$. Because the relative humidities are the same, the entropy generation is approximately zero (a small amount of entropy is generated in the cycle's heat exchanger). Hence the traditional desiccant cycle COP is approximately the Carnot COP at this minimum possible regeneration temperature. For the inlet and outlet conditions in Table 1, the minimum regeneration temperature was found to be 55.3° C. At this minimum, the effective source temperature of the cycle (T_s) is low, which is undesirable, but the entropy generation is nearly zero, which is desirable.

[0122] In practical operation, traditional desiccants must be regenerated with temperatures greater than the theoretical minimum. As the regeneration temperature increases, the relative humidity of the regeneration air decreases. A regeneration temperature greater than the minimum creates a mismatch in relative humidity (and, correspondingly, a mismatch in chemical potential) between the process outlet air and regeneration inlet air. This increases the rate of desorption required to operate a practical, finitely sized system. It also results in mass transfer of water vapor across a non-zero chemical potential difference, which results in entropy generation. Thus, for high temperature regeneration sources, the traditional desiccant cycle COP becomes limited by entropy generation and cannot reach the Carnot COP for that source temperature. Because the LCST cycle is

regenerated with a phase transition, as opposed to dry air, it does not suffer from the entropy generation that hinders the traditional desiccant cycle at higher regeneration temperatures.

[0123] It is important to note that the COP values presented in FIG. 7 and Table 1 are the maximum possible COPs that both the LCST and traditional desiccant cycles could achieve for the given operating conditions.

[0124] In summary, a thermoresponsive polymer desiccant cooling cycle that utilizes LCST polymer gels is presented. This example provides the thermodynamic equations necessary to analyze the performance of this cycle both for the reversible limit and for practical, irreversible operation. The LCST cycle can significantly outperform traditional desiccants that require high temperature heat sources. Generally, the cycle COP improves as the gel transition temperature increases, until sensible heating begins to dominate. The sensible heating can be mitigated by using a highly effective recuperator or a gel with a high swelling ratio. When used with higher temperature heat sources, the LCST cycle can operate more efficiently than traditional desiccant cycles. On the other hand, the LCST cycle is not constrained by the minimum regeneration temperature experienced in traditional desiccant cycles. This allows the LCST cycle to use lower regeneration temperatures and more readily use “free” heat sources like solar- or waste-heat, thereby reducing electricity consumption.

[0125] Analysis of the LCST cycle for reversible swelling is achieved by setting entropy generation to zero and solving for the enthalpy and entropy of swelling. When considering a real polymer, the enthalpy and entropy of swelling can be measured and will be greater in magnitude than the reversible values. This behavior results from the chemical potential of water within the polymer being lower than the water vapor in the air, an irreversible behavior that is necessary to drive the real swelling process. To analyze the cycle for irreversible (realistic) swelling, Equation 2 can be used with potential future data on LCST polymers in humid air. For now, the analysis in this article serve as an upper limit on performance.

[0126] Aside from entropy generation during swelling, real thermoresponsive polymers present several other non-ideal behaviors not captured in this analysis. One example of non-ideal behavior is seen in the volume phase transition of thermoresponsive gels. This phase transition is an abrupt change in the polymer volume fraction (or water content) that occurs upon heating at a certain temperature. When an LCST gel reaches the transition temperature, any addition of heat will not cause the temperature to increase but will cause the swelled phase to transition to a shrunken phase and liquid water phase. This will continue until the gel is entirely in the shrunken phase, making the phase transition an isothermal one.

[0127] FIG. 11 illustrates an example LCST cycle wherein a polymer swells in humid air, thereby dehumidifying the air. Next it is heated to expel the absorbed water. Finally, it is cooled back down to start the cycle over.

[0128] FIG. 12 illustrates an example layout of a LCST cycle cooling system wherein humid air is isothermally dehumidified by a swelling polymer and is then sensibly cooled in an indirect evaporative cooler. Solar heat is used to regenerate the polymer. Expelled liquid water is recovered

for use in the evaporative cooler. Since the heat input comes from the sun, the only electrical power input is at the fans and pumps.

[0129] FIG. 13 illustrates an example polymer that has a transition temperature of -32°C . (-90°F). The polymer swells due to absorption of water below this temperature (left photo). The polymer shrinks due to desorption of water above this temperature (right photo).

[0130] It is to be understood that the embodiments and claims disclosed herein are not limited in their application to the details of construction and arrangement of the components set forth in the description and illustrated in the drawings. Rather, the description and the drawings provide examples of the embodiments envisioned. The embodiments and claims disclosed herein are further capable of other embodiments and of being practiced and carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein are for the purposes of description and should not be regarded as limiting the claims.

[0131] Accordingly, those skilled in the art will appreciate that the conception upon which the application and claims are based may be readily utilized as a basis for the design of other structures, methods, and systems for carrying out the several purposes of the embodiments and claims presented in this application. It is important, therefore, that the claims be regarded as including such equivalent constructions.

[0132] Furthermore, the purpose of the foregoing Abstract is to enable the United States Patent and Trademark Office and the public generally, and especially including the practitioners in the art who are not familiar with patent and legal terms or phraseology, to determine quickly from a cursory inspection the nature and essence of the technical disclosure of the application. The Abstract is neither intended to define the claims of the application, nor is it intended to be limiting to the scope of the claims in any way.

What is claimed is:

1. An air conditioning process comprising:

a latent cooling stage comprising:

absorbing, via a responsive hygroscopic material, moisture from ambient air;

heating, via a heat source, the responsive hygroscopic material above a transition temperature wherein the responsive hygroscopic material transitions from being hygroscopic to being hydrophobic;

expelling, from the responsive hygroscopic material, liquid; and

sensible cooling the responsive hygroscopic material below the transition temperature.

2. The process of claim 1 further comprising:

a sensible cooling stage comprising:

sensible cooling, via an evaporative cooler, the ambient air,

wherein the evaporative cooler evaporates the liquid expelled from the responsive hygroscopic material.

3. The process of claim 2, wherein the evaporative cooler is one of a direct evaporative cooler, indirect evaporative cooler, or a partially direct and partially indirect evaporative cooler.

4. The process of claim 2, wherein the latent and sensible cooling of the ambient air is performed without the use of conventional refrigerants.

5. The process of claim 2 further comprising:
moving the responsive hygroscopic material from a first position to a second position,
wherein at the first position, the absorbing, via a responsive hygroscopic material, moisture from ambient air step is performed, and
wherein at the second position, the heating, via the heat source, the responsive hygroscopic material above a transition temperature step is performed.
6. The process of claim 1 further comprising:
transferring at least a portion of the heat removed from the hygroscopic material by sensible cooling the responsive hygroscopic material below the transition temperature;
storing the heat in a recuperator;
and transferring at least a portion of the stored heat back to the responsive hygroscopic material.
7. The process of claim 1, wherein the responsive hygroscopic material comprises PNIPAAm (poly(N-isopropylacrylamide)) and one or more hydrogels.
8. The process of claim 1, wherein the heat source comprises one or more of solar heat, waste heat, gas heat, or electric heat.
9. An air conditioning system comprising:
a latent cooling stage comprising:
a responsive hygroscopic material configured to absorb moisture from ambient air;
a heat source configured to heat the responsive hygroscopic material above a transition temperature wherein the responsive hygroscopic material transitions from being hygroscopic to being hydrophobic causing the responsive hygroscopic material to expel a liquid previously absorbed from the ambient air; and
a heat exchanger configured for sensible cooling the responsive hygroscopic material.
10. The system of claim 9 further comprising:
a sensible cooling stage comprising:
an evaporative cooler configured for sensible cooling the ambient by evaporating the liquid expelled from the responsive hygroscopic material.
11. The system of claim 10, wherein the evaporative cooler is one of a direct evaporative cooler, indirect evaporative cooler, or a partially direct and partially indirect evaporative cooler.
12. The system of claim 10, wherein the latent and sensible cooling of the ambient air is performed without the use of conventional refrigerants.
13. The system of claim 9 further comprising:
an actuator configured to move the responsive hygroscopic material from a first position to a second position,
wherein at the first position, the responsive hygroscopic material absorbs moisture from the ambient air, and

wherein at the second position, a heat source heats the responsive hygroscopic material above the transition temperature.

14. The system of claim 9 further comprising:
a first polymer bed comprising the responsive hygroscopic material;
a second polymer bed comprising the responsive hygroscopic material; and
one or more actuators configured to change the airflow, as between inside air and outside air, passing over each of the first polymer bed and second polymer bed.

15. The system of claim 9 further comprising:
a recuperator configured to transfer at least a portion of the heat removed from the hygroscopic material by sensible cooling the responsive hygroscopic material below the transition temperature, storing the heat in the recuperator, and transferring at least a portion of the stored heat back to the responsive hygroscopic material.

16. The system of claim 9, wherein the responsive hygroscopic material comprises PNIPAAm (poly(N-isopropylacrylamide)) and one or more hydrogels.

17. The system of claim 9, wherein the heat source comprises one or more of solar heat, waste heat, gas heat, or electric heat.

18. The system of claim 9, wherein the heat exchanger is an ambient heat exchanger.

19. An air conditioning system comprising:

- a latent cooling stage comprising:
a responsive hygroscopic material configured to absorb moisture from ambient air;
a heat source configured to heat the responsive hygroscopic material above a transition temperature wherein the responsive hygroscopic material transitions from being hygroscopic to being hydrophobic causing the responsive hygroscopic material to expel a liquid previously absorbed from the ambient air; and
a heat exchanger configured for sensible cooling the responsive hygroscopic material; and
a sensible cooling stage comprising:
an evaporative cooler configured for sensible cooling the ambient by evaporating the liquid expelled from the responsive hygroscopic material.

20. The system of claim 19 further comprising:
a recuperator configured to transfer at least a portion of the heat removed from the hygroscopic material by sensible cooling the responsive hygroscopic material below the transition temperature, storing the heat in the recuperator, and transferring at least a portion of the stored heat back to the responsive hygroscopic material.

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